
SyMerge: From Non-Interference to Synergistic Merging via Single-Layer Adaptation

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Abstract

Model merging combines independently trained models into a single multi-task model. However, most existing approaches focus primarily on avoiding task interference. We argue that its greater potential lies in enabling task synergy, where tasks actively improve one another. We identify cross-task performance, defined by compatibility between encoders and predictors across tasks, as a key indicator of merge quality. We demonstrate that adapting only a single task-specific layer is sufficient to induce such synergy. We propose SyMerge, a lightweight framework that jointly optimizes merging coefficients and a single task-specific layer. We adopt an expert-guided self-labeling objective, providing stable supervision beyond entropy minimization. Intriguingly, we further show that SyMerge successfully merges models trained from different initializations, a regime where standard methods break down. Our minimalist yet principled method achieves state-of-the-art results across vision, dense prediction, and NLP benchmarks. Our code is available at <https://aim-skku.github.io/SyMerge>.

1. Introduction

Multi-task learning (Caruana, 1997) solves multiple tasks within a single model but requires joint training and careful balancing (Yu et al., 2020; Misra et al., 2016; Kendall et al., 2018; Hu et al., 2023). Model merging (Wortsman et al., 2022a; Ilharco et al., 2022) instead combines independently fine-tuned models at the parameter level using task vectors (Ilharco et al., 2023). However, naively aggregating task vectors often leads to severe performance degradation

due to task interference (Yadav et al., 2023; Gargiulo et al., 2025; Wang et al., 2024). This motivates test-time adaptive approaches that tune merged models in an unsupervised manner (Yang et al., 2024b; Tang et al., 2024a; Yang et al., 2024a; Wei et al., 2025), though it remains unclear whether such methods fundamentally resolve task interference.

To address task interference more explicitly, prior work has explored a range of mitigation strategies, including parameter-level masking or pruning (Wang et al., 2024; Yadav et al., 2023; Du et al., 2024; Deep et al., 2024), structural analysis of weight matrices via SVD (Gargiulo et al., 2025; Marczak et al., 2025; Panariello et al., 2025), and frequency-based filtering (Zheng & Wang, 2025). Other approaches pursue non-interference through weight disentanglement, aiming to ensure that adding one task vector does not impair another (Ortiz-Jimenez et al., 2023; Jin et al., 2025). Despite these advances, most existing methods continue to treat non-interference as the final objective.

In this paper, we argue that model merging should go beyond avoiding interference and explicitly consider cross-task interactions. The goal is to achieve *synergy*, where tasks actively benefit one another. We define synergy not merely as improved average performance, but as representation-level realignment: the merged encoder becomes compatible with task-specific predictors across tasks, enabling cross-task mappings unattainable by any individual encoder. This view is supported by a pilot study showing that cross-task performance, obtained by pairing a classifier from one task with an encoder from another, strongly predicts merge quality. We further observe that adapting even a single task-specific layer can greatly improve cross-task compatibility, suggesting that minimal adaptation can be key to inducing synergy in merging.

Building on this intuition, we propose **SyMerge**, a lightweight and test-time adaptive framework that jointly optimizes a single task-specific layer and the encoder’s merging coefficients. Unlike conventional entropy minimization (Yang et al., 2024b; Tang et al., 2024a), which we find unstable, we employ a robust self-labeling strategy guided by expert model predictions. As adaptation progresses, entropy minimization can drift away from the supervised cross-entropy objective, whereas ours stays strongly aligned and

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provides a stable training signal. This design yields strong generalization under distribution shifts without introducing extra modules or costly training. This setup aligns with recent test-time adaptation studies for robust distribution-shift handling (Liang et al., 2025; Zhang et al., 2025c;b). Finally, **SyMerge** also works when merging models trained from different initializations, where existing merging methods usually fail. Our contributions are threefold:

- We redefine the goal of model merging from mitigating interference to fostering positive task synergy, moving beyond non-interference or disentanglement.
- We present **SyMerge**, a minimalist yet effective method that adapts only a single layer and merging coefficients, stabilized by expert-guided self-labeling.
- **SyMerge** achieves state-of-the-art performance across vision (classification), dense prediction (segmentation, depth/surface normal estimation), and NLP tasks.

2. Background

2.1. Preliminary

Problem definition. Multi-Task Learning (MTL) aims to train a single model that performs well across a set of K tasks. Model merging offers an efficient alternative by constructing a multi-task model without costly joint training (Kim et al., 2025). A pre-trained model Θ_{pre} is fine-tuned independently on each task, producing K expert models with weights $\Theta_1, \dots, \Theta_K$. The key challenge is to design a merging function $\mathcal{M}$ that combines these experts into a unified parameter set $\Theta_{\text{MTL}} = \mathcal{M}(\Theta_1, \dots, \Theta_K)$. The merged model has a shared encoder and task-specific layers. For task k , the model can be written as $f(\cdot; \Theta_k) = f^L(\cdot; \theta_k^L) \circ f^{1:L-1}(\cdot; \Theta_{\text{MTL}}^{\text{enc}})$, where $\Theta_{\text{MTL}}^{\text{enc}}$ denotes the merged encoder. The goal is to ensure that this single model f achieves strong performance across all K tasks simultaneously.

Training-free model merging. These methods determine merging coefficients using predefined heuristics (Yadav et al., 2023; Du et al., 2024) or through costly hyperparameter grid search on a validation set (Wang et al., 2024; Ilharco et al., 2023). While this avoids data-driven optimization on the target distribution, it introduces critical scalability and computational bottlenecks. The requisite grid search becomes computationally prohibitive as evaluation cost scales with the number of tasks, and the memory-intensive element-wise operations often preclude GPU acceleration. A representative approach is *Task Arithmetic* (Ilharco et al., 2023), which captures the task-specific knowledge as the deviation from the pre-trained weights, $\tau_k^l = \theta_k^l - \theta_{\text{pre}}^l$. The merging process is applied to the encoder layers ($l \in \{1, \dots, L-1\}$), where the merged weights are constructed as $\theta_{\text{MTL}}^l = \theta_{\text{pre}}^l + \lambda \cdot \sum_{k=1}^K \tau_k^l$. Here,

λ is a hyperparameter. The final multi-task model is then composed of this single merged encoder and the original task-specific layers from their respective fine-tuned models. However, while simple, their data-agnostic nature incurs suboptimal performance compared to adaptive methods, especially under many-task settings or distribution shifts.

Test-time adaptive merging. Another line of methods adapts to the target distribution by optimizing parameters using unlabeled test data. Since ground-truth labels are unavailable, these methods rely on proxy objectives such as minimizing prediction entropy (Yang et al., 2024b; Tang et al., 2024a). However, entropy-based optimization is limited to classification and does not easily extend to tasks like semantic textual similarity or dense per-pixel prediction. Post-hoc calibration offers an alternative that avoids this issue. It adjusts the feature representations of a merged model using additional adapters, rather than learning the merging coefficients directly (Yang et al., 2024a; Wei et al., 2025).

A milestone work is *AdaMerging* (Yang et al., 2024b) that improves *Task Arithmetic* by constructing a merged model $f(x; \Theta_{\text{MTL}})$ through learning merging coefficients, denoted by $\Lambda = \{\lambda_k^l\}_{k,l}$, in a task-/layer-wise manner. The merged weights are defined as $\theta_{\text{MTL}}^l = \theta_{\text{pre}}^l + \sum_{k=1}^K \lambda_k^l \cdot \tau_k^l$. The coefficients Λ are learned by optimizing the following objective function: $\min_{\Lambda} \sum_{k=1}^K \mathbb{E}_{x \in \mathcal{X}_k^{\text{te}}} [\mathcal{L}_k(f(x; \Theta_{\text{MTL}}))]$. Note that ground-truth test labels are not used; it learns coefficients via entropy minimization loss $\mathcal{L}_k$. Our method builds upon test-time merging approaches. The following section motivates our method and outlines our future directions.

2.2. Motivation

Limitations of training-free methods. Our first motivation arises from the limitation of training-free methods, which are vulnerable when adapting to unseen tasks or domain shifts. To examine whether existing merging methods lack robustness to data quality and distribution shifts, we intentionally corrupt 4 task datasets following Hendrycks & Dietterich (2019) as a prevalent real-world challenge. This setup serves as a rigorous proxy for noisy or out-of-distribution test sets often encountered in the wild.

As shown in Figure 1, the performance of training-free methods drops significantly on this corrupted data, revealing their fundamental limitation in handling domain shifts. This failure motivates our approach of leveraging unlabeled test data for robust merging, a principle adapted from test-time adaptation that avoids data leakage. Our empirical results are consistent with prior theoretical work (Xu et al., 2025), which highlights the vulnerability of data-agnostic merging in practice. For more detailed results, see Appendix C.1.

Rethinking cross-task performance. Another motivation comes from the intuition that a model’s cross-task perfor-

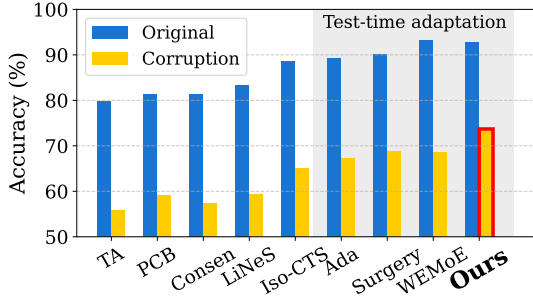


Figure 1. **Training-Free methods collapse under corruption.** Worse than test-time methods on clean data as well, and far more degraded under corruption.

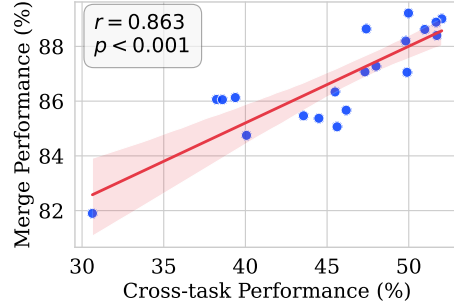


Figure 2. **Cross-task vs. Merge Performance.** Positive correlation observed across 20 vision tasks with regression fit and 95% confidence interval.

mance¹ is closely tied to its merging performance. To examine this, we conducted a preliminary study on 20 vision tasks using ViT-B/32. We observed a significant positive correlation ($r = 0.863$, $p < 0.001$) between a model’s average cross-task performance and its average merging performance. Specifically, we treat each task in the benchmark as a target task B and iterate through all other tasks A ($A \neq B$). For each pair, we evaluate: (1) the pairwise cross-task performance, and (2) the pairwise merging performance by weight-averaging the two encoders and then evaluating the new merged encoder on B ’s task. The final metrics for task B are computed by averaging these pairwise values over all source tasks A . As shown in Figure 2, a model’s ability to generalize across tasks is a strong and reliable predictor of its potential for successful merging. This key insight forms the foundation of our work.

Prior work (Ortiz-Jimenez et al., 2023) often regards non-interference as the objective of model merging, where the sole aim is to prevent tasks from degrading one another. However, this perspective inherently imposes a ceiling because cross-task performance cannot go beyond that of the pre-trained model, leaving little room for improvement. We instead highlight the need for positive synergy, where tasks actively enhance one another and collectively achieve performance beyond this ceiling. In Section 3.2, we provide a theoretical analysis showing that improvements in cross-task performance directly translate into more effective merging.

3. Method

Our method uses the link between cross-task performance and merge quality as a structural motivation. To study this connection, we compare a baseline that pairs a task’s original classifier—over-specialized to its native encoder—with an encoder from another task. For each coefficient value, we construct a fixed merged encoder using Task Arithmetic (Ilharco et al., 2023) in the 8-task ViT-B/32 setting

¹We define *cross-task performance* as evaluating model A ’s encoder with model B ’s classifier on B ’s task.

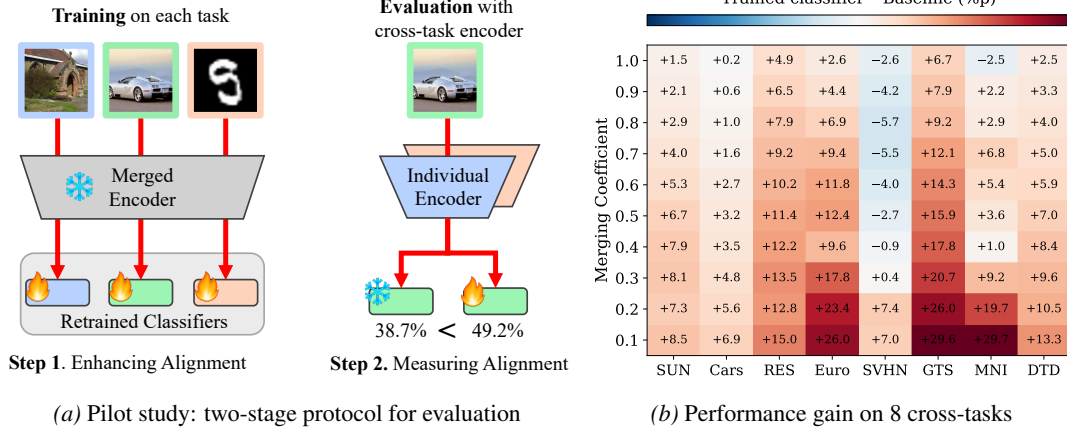
under different merging coefficients. We then design a two-stage protocol (Figure 3a): first retraining the task-specific layer $f^L(\cdot; \theta_k^L)$ on features from a fixed merged encoder $f^{1:L-1}(\cdot; \Theta_{\text{MTL}}^{\text{enc}})$ using labeled data $\mathcal{X}_k^{\text{tr}}$, and then evaluating the updated layer on an encoder from a different task $m \neq k$. This procedure provides a direct measure of functional alignment (Csiszárík et al., 2021) across tasks.

As shown in Figure 3b, the protocol consistently improves cross-task performance over the baseline (detailed results are in Appendix C.2). While the benefit is observed in most cases, SVHN displays limited or negative gains, likely due to its overlap with MNIST. We also find that the effect is not restricted to the final classifier: adapting a single intermediate block yielded similar improvements (Figure 6). Finally, the improvement relies on the merging coefficients, highlighting the importance of a well-formed merged encoder for stronger alignment. Together with Figure 2 and Section 3.2, these results suggest that single-layer adaptation can improve cross-task compatibility in a controlled frozen-encoder setting and thereby benefit merging. This connection is used to motivate the design of our method, rather than as an explicit optimization objective. However, this protocol relies on ground-truth labels that are unavailable during model merging. The remaining challenge is therefore to achieve the same effect in an unsupervised setting, which is the focus of our proposed method, **SyMerge**.

3.1. SyMerge: Synergistic Model Merging

SyMerge addresses the lack of labels at test time by adopting a self-labeling strategy. A common approach in this setting is to use entropy minimization as a supervisory signal (Yang et al., 2024b). However, this proxy can be unstable and misguide the model, as shown in our analysis (Figure 4) and by prior work (Oh et al., 2025).

To create a more reliable training signal, we use the individually fine-tuned models as expert teachers and adopt a self-labeling strategy (Lee et al., 2013). Our objective is to train the merged model to match the confident predic-



(a) Pilot study: two-stage protocol for evaluation

(b) Performance gain on 8 cross-tasks

Figure 3. Pilot study protocol and its results on 8 cross-tasks using ViT-B/32. (a) We first enhance a classifier’s functional alignment by training it on representations from a general-purpose merged encoder. We then measure this enhancement by evaluating the trained classifier’s cross-task performance when paired with the encoder of a different, individual task. (b) The heatmap shows the accuracy gain (%p) over the baseline across 8 tasks (x-axis) under various merged encoder configurations (y-axis, via merging coefficient). The consistently positive gains (red) demonstrate the protocol’s effectiveness in enhancing functional alignment.

tions of these expert models. We formulate this as minimizing the cross-entropy between the predictions of our merged model and the self-labels from the individual models on the unlabeled test set $\mathcal{X}^{te}$. The objective is defined

as $\min_{\{\lambda_k^l\}, \{\theta_k^{tr}\}} \sum_{k=1}^K \mathcal{L}_{CE} (C_k^{\text{merged}}, C_k^{\text{ft}})$, where (i) C_k^{merged} is

the output from our merged model for task k ; (ii) C_k^{ft} is the fixed prediction from the corresponding expert model; and (iii) the trainable parameters are the merging coefficients $\{\lambda_k^l\}$ and a task-specific layer $\{\theta_k^{tr}\}$. Refer to Appendix E.2 for details. Unlike conventional teacher-student distillation, which trains a separate student model, **SyMerge** directly optimizes the merging coefficients and a single task-specific layer, thereby promoting functional alignment within the merged model. Our objective readily extends beyond classification, as the key principle is to mimic the expert’s output signal. This is achieved simply by using the appropriate task-specific loss (e.g., L1 loss for regression tasks). Below, we provide an empirical justification for choosing this self-labeling objective over the more common, but less stable, entropy minimization.

Note that a key difference from AdaMerging (Yang et al., 2024b) is that we jointly optimize both the shared encoder’s merging coefficients and the task-specific layer. We argue that this co-adaptation improves the merged encoder itself (not merely the chosen adaptation layer) and enables stable recovery even in disjoint-basin regimes where coefficient-only tuning often collapses (refer to Section 4.3). We can optionally apply a confidence-based filtering mechanism to further enhance performance (see details in Appendix B).

Our objective function. An effective proxy objective for test-time adaptation must maintain a strong correlation with

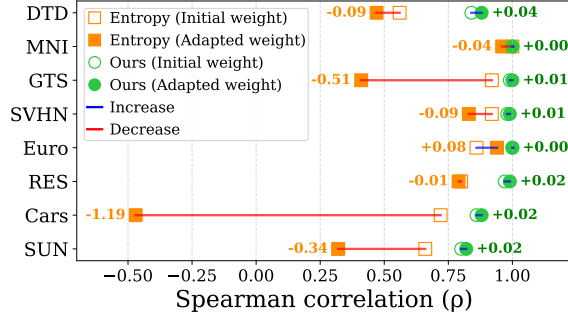


Figure 4. Correlation of proxy losses with the ground truth loss. The ground truth loss is defined as the supervised cross-entropy computed with ground truth labels. We compare coefficients for Entropy and Ours using merged weights before and after training. A coefficient closer to +1 indicates a more reliable proxy.

the ground-truth objective. To verify this, we compute the Spearman correlation (a rank-based measure of monotonic association) on 8 vision tasks with ViT-B/32. A higher correlation indicates that the loss function provides a reliable training signal. We evaluate this correlation for both the initial model merged via Task Arithmetic (“Initial weight”) and the final model after it has been optimized with a proxy loss (“Adapted weight”). For the “Entropy”, we correlate the entropy of the model’s output with the ground-truth loss. In contrast, for “Ours”, we correlate our self-labeling loss with the same ground-truth loss.

Figure 4 shows that entropy initially correlates moderately with supervised cross-entropy. However, after training, this correlation significantly weakens, and in some cases, entropy diverges sharply (e.g., Cars). This suggests entropy is not a stable proxy for supervision and may fail to maintain consistency as training progresses, which aligns with prior studies (Yang et al., 2024b; Oh et al., 2025). In contrast, our

Table 1. Multi-task performance across 8, 14, and 20 vision tasks. We compare our method with the competing methods for merging ViT-B/32 and ViT-L/14 fine-tuned models. Our main competitors are the test-time merging methods (bottom group, starting from AdaMerging). All reported results for our methods are the mean and standard deviation computed over 5 runs.

Method	ViT-B/32			ViT-L/14		
	8 tasks	14 tasks	20 tasks	8 tasks	14 tasks	20 tasks
Individual	90.5	89.5	90.4	94.2	93.2	94.0
Task Arithmetic (Ilharco et al., 2023)	69.1	65.4	60.8	84.5	78.4	73.9
Ties-Merging (Yadav et al., 2023)	72.9	65.2	63.1	86.0	80.5	73.0
PCB Merging (Du et al., 2024)	75.6	63.8	52.7	87.5	81.3	73.4
LiNeS w/ TA (Wang et al., 2025)	74.1	68.0	63.7	86.9	80.4	75.7
Consensus TA (Wang et al., 2024)	74.9	70.3	65.0	86.6	82.3	78.9
TSV-M (Gargiulo et al., 2025)	84.0	80.1	76.6	91.5	88.2	87.2
ISO-Merging (Marczak et al., 2025)	84.2	80.6	76.9	93.0	89.7	89.2
EMR Merging (Huang et al., 2024a)	88.7	86.1	86.6	93.7	91.4	92.0
LOT Merging (Sun et al., 2025)	82.7	77.3	73.1	90.5	86.2	84.1
AdaMerging (Yang et al., 2024b)	80.1	76.7	69.6	90.8	85.2	82.1
Surgery w/ Ada. (Yang et al., 2024a)	87.5	84.6	84.5	92.3	90.4	89.5
WEMoE (Tang et al., 2024a)	89.4	83.0	78.9	93.6	88.4	78.8
ProbSurgery w/ Ada. (Wei et al., 2025)	87.4	84.9	84.5	92.7	90.7	90.2
SyMerge	90.1 ± 0.1	88.7 ± 0.1	88.6 ± 0.4	94.1 ± 0.0	92.8 ± 0.1	93.2 ± 0.1

proposed objective maintains a consistently high correlation, providing a stable and reliable supervisory signal.

3.2. Theoretical Backup

We theoretically support that stronger cross-task performance leads to better merge performance. Following Ortiz-Jimenez et al. (2023), non-interference (*i.e.*, weight disentanglement) assumes that adding a task vector τ_i leaves another task j unchanged: $L_j[f(x; \theta_0 + \tau_i)] = L_j[f(x; \theta_0)]$ for all $i \neq j$. In contrast, synergy requires a stronger condition, where τ_i improves task j : $L_j[f(x; \theta_0 + \tau_i)] < L_j[f(x; \theta_0)]$.

Proposition 3.1. *Assume cross-task linearity (Zhou et al., 2024) so that $f(x; \frac{1}{2}(\theta_i + \theta_j)) \approx \frac{1}{2}f(x; \theta_i) + \frac{1}{2}f(x; \theta_j)$, and suppose the loss function is convex in its output. Then the merged model $f_{\text{merge}}(x) = f(x; \frac{1}{2}(\theta_i + \theta_j))$ has an expected loss below the case of weight disentanglement.*

The proof in Appendix D shows that this positive cross-task improvement tightens the convex upper bound, supporting cross-task performance as a structural indicator of merge quality rather than as the direct optimization objective of SyMerge. We note that convexity is required only with respect to the model outputs $f(x; \theta)$, not the parameters θ , and holds exactly for cross-entropy loss. Although cross-task linearity is an approximation in parameter space, Appendix E.4 empirically verifies that it holds robustly in our setting.

4. Experiment

4.1. Experimental Setup

Datasets and metrics. We follow the standard setups in Ilharco et al. (2023), using fine-tuned weights on 8 image

classification tasks, and extend it to 20 tasks following Wang et al. (2024). We report accuracy for all vision tasks. For dense prediction, we use NYUv2 (Silberman et al., 2012), containing 1,449 RGB-D images. It includes three tasks: 13-class semantic segmentation, depth estimation, and surface normal estimation. We evaluate them with mIoU/pixel accuracy, absolute/relative error, and mean angular error, respectively. For NLP tasks, we adopt the setups from Yu et al. (2024) and Huang et al. (2024a), using fine-tuned weights for 8 GLUE tasks (Wang et al., 2019). Task-specific metrics are employed: Matthews correlation for CoLA (Warstadt et al., 2019), Pearson/Spearman correlations for STS-B (Cer et al., 2017), and accuracy for others.

Models. For vision experiments, we employ pre-trained CLIP (Radford et al., 2021) models, specifically ViT-B/32, L/14. Most analyses use ViT-B/32. To ensure comparability across 8 tasks, we use publicly available fine-tuned weights from Ilharco et al. (2023). For 14- and 20-task experiments, we fine-tune CLIP ViT-B/32 following the Task Arithmetic configuration. For dense prediction, we adopt ResNet-50 (He et al., 2016) as the backbone, following Tang et al. (2025). We fine-tune ImageNet (Deng et al., 2009) pre-trained models for each task. For NLP tasks, we use RoBERTa-base (Liu et al., 2019), leveraging publicly available fine-tuned weights from Huang et al. (2024a).

4.2. Main Results

Merging 8, 14, 20 vision tasks. Table 1 demonstrates the superiority of our approach, surpassing all baselines across varying numbers of tasks and model scales. Notably, while the performance of competing methods degrades sharply with more tasks, SyMerge remains consistent and stays

Table 2. **Multi-task performance across three dense prediction tasks.** We compare the performance of merging ResNet-50 models fine-tuned on three dense prediction tasks in NYUv2.

Method	Segmentation		Depth Estimation		Normal Mean↓
	mIoU↑	Pix Acc↑	Abs Err↓	Rel Err↓	
Individual	52.0	74.2	41.5	17.3	24.2
Weight Averaging	36.6	64.0	55.0	23.2	30.0
Task Arithmetic (Ilharco et al., 2023)	31.6	60.3	56.7	24.0	30.6
Ties-Merging (Yadav et al., 2023)	39.9	62.7	61.3	27.3	36.2
MagMax (Marczak et al., 2024)	24.7	54.7	60.3	23.9	30.3
LiNeS w/ TA (Wang et al., 2025)	36.2	64.0	54.2	22.4	29.1
EMR-Merging (Huang et al., 2024a)	41.5	67.2	48.6	19.4	26.5
Surgery w/ TA (Yang et al., 2024a)	43.3	67.4	55.3	24.7	34.7
ProbSurgery w/ TA (Wei et al., 2025)	43.6	67.6	52.6	22.3	36.7
SyMerge	49.8 ±0.3	73.1 ±0.2	45.3 ±0.6	18.8 ±0.5	26.2 ±0.1

Table 3. **Multi-task performance across 8 NLP tasks.** We present a performance comparison of merging the RoBERTa models fine-tuned on 8 tasks in GLUE.

Method	CoLA	SST2	MRPC	STS B	QQP	MNLI	QNLI	RTE	Avg.
Individual	60.2	94.0	89.2	90.6	91.4	87.2	92.7	79.1	85.6
Weight Averaging	14.0	64.1	69.4	31.8	75.4	42.2	58.7	55.2	51.3
Task Arithmetic	18.8	85.9	79.9	74.0	83.8	59.1	69.7	62.1	66.7
MagMax	17.3	76.0	70.8	71.3	85.8	70.4	59.5	45.1	62.0
Ties-Merging	20.5	84.4	81.1	58.2	85.7	64.7	74.8	43.0	64.0
LiNeS	26.1	86.4	78.9	72.9	83.3	56.0	75.9	59.9	67.4
TSV-M	28.7	88.4	83.1	76.8	85.8	59.9	75.2	47.7	68.2
ISO-Merging	27.3	86.2	79.2	73.7	85.9	71.7	81.1	70.4	71.9
EMR-Merging	40.0	93.4	86.3	82.8	89.7	85.5	89.6	74.4	80.2
Surgery w/TA	46.8	93.6	89.7	88.7	85.1	78.0	85.3	76.5	80.5
ProbSurgery w/TA	54.7	93.6	89.7	89.4	85.2	77.6	85.3	76.9	81.6
SyMerge	60.0 ±0.1	93.8 ±0.3	89.2 ±0.0	90.4 ±0.2	85.2 ±1.1	84.0 ±0.4	89.2 ±0.5	79.1 ±0.0	83.9 ±0.2

close to the single-task expert reference. Furthermore, it establishes a large margin over other test-time adaptive methods. Per-task performance can be found in Appendix C.3.

Merging dense prediction tasks. Dense prediction tasks (Zhang et al., 2022; 2025a), such as segmentation (Choi et al., 2023), depth estimation (Min et al., 2025), and normal estimation (Li et al., 2015), pose significant challenges due to their inter-task heterogeneity; for example, semantic segmentation relies on high-level semantic abstraction, whereas depth and surface normal estimation depend on low-level geometric features. Despite this, most model merging studies focus on classification, leaving these high-interference scenarios largely unexplored. Table 2 presents results for these tasks, highlighting the limitations of previous methods. ProbSurgery, which performs comparably to individual models on 8 classification tasks, suffers substantial performance drops in depth and normal estimation. Similarly, EMR-Merging, despite leveraging task-specific dynamic masks, struggles to maintain segmentation performance. In contrast, our approach effectively balances these tasks, maintaining performance close to individual models even in dense prediction.

Merging 8 NLP tasks. As shown in Table 3, **SyMerge** substantially outperforms the best method on average. A key observation comes from CoLA, which evaluates grammatical acceptability and differs from other, more semantic GLUE tasks. Due to this task heterogeneity, distillation-based methods (*e.g.*, Surgery, ProbSurgery) and EMR-Merging suffer from a significant performance drop on CoLA, whereas **SyMerge** remains close to individual models across all tasks. These results highlight our method’s robustness, as it not only maximizes average performance but also prevents significant degradation on any single task, enabling stable integration of diverse tasks.

4.3. Empirical Analyses

Transferable cross-task ability. Our work argues that enhancing functional alignment is key to successful merging, and our pilot study showed that adapting a single layer is an effective way to achieve this. We now investigate if the adapted layer learns a truly transferable capability, *i.e.*, if it can be reused as a plug-and-play component to improve other merging methods. To test this, we use the merged encoders from two leading baselines, Task Arithmetic (Il-

Table 4. Cross-task transferability check. Classifiers trained with our method replace the original zero-shot classifiers (in a pre-trained model) connected to merged encoders (Task Arithmetic and AdaMerging) for evaluating different tasks. This replacement yields substantial performance gains on both merged and cross-task evaluations without training on target tasks, demonstrating the high transferability and improved functional alignment.

Encoder	Classifier	Merged	Cross
Task Arithmetic	Zero-shot	69.1	49.8
	SyMerge	79.6 (+10.5)	54.8 (+5.0)
AdaMerging	Zero-shot	80.1	50.1
	SyMerge	87.7 (+7.6)	54.1 (+4.0)

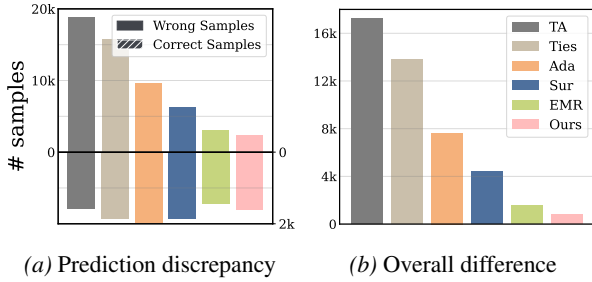


Figure 5. Prediction discrepancies between merged and individual models. (a) Upper bars indicate cases correctly predicted by individual models but missed by the merged model; lower hatched bars indicate the opposite. (b) Overall difference.

harco et al., 2023) and AdaMerging (Yang et al., 2024b), as fixed backbones. We then replace their original zero-shot classification heads with the corresponding classifiers trained by our **SyMerge** process and evaluate the new model combinations without any further training.

In Table 4, we assess this transferability using two metrics. “Merged” indicates the accuracy of the final multi-task model, which combines a baseline’s merged encoder (*i.e.*, $\theta_{pre}^l + \sum_k \lambda_k^l \cdot \tau_k^l$) with the classifier. “Cross”, on the other hand, quantifies average generalization by pairing an encoder built from a single scaled task vector for task i (*i.e.*, $\theta_{pre}^l + \lambda_i^l \cdot \tau_i^l$) with a classifier from a different task j (where $i \neq j$) and averaging the results over all such pairs.

The results show a remarkable degree of transferability. When paired with the TA encoder, it boosts merged performance by 10.5%p and, critically, improves cross-task performance by 5.0%p. We observe similar substantial gains with the AdaMerging encoder. These significant gains demonstrate that the layer trained by **SyMerge** learns a robust and general function not overfitted to its original encoder. This provides strong evidence that our method effectively enhances functional alignment and produces highly transferable components.

Positive/negative transfer. To evaluate the synergistic effect of **SyMerge**, we analyze prediction discrepancies between the merged model and the individual models. Fig-

Table 5. Merging models with different initializations. We merge EuroSAT (DataComp-ViT) and DTD (OpenAI-ViT) from disjoint basins. While standard methods collapse due to severe loss barriers, **SyMerge** uniquely recovers performance, showing its capability to bridge the gap where linear connectivity fails.

Method	EuroSAT	DTD	Avg.
Individual (EuroSAT)	98.1	3.6	50.8
Individual (DTD)	2.0	79.4	40.7
Weight Averaging	11.6	2.2	6.9
AdaMerging	8.6	2.1	5.4
Surgery	38.1	13.1	25.6
SyMerge	96.2 (+58.1)	62.0 (+48.9)	79.1 (+53.5)

ure 5a visualizes this by distinguishing negative transfer (upper bars: merged model fails, individual succeeds) from positive transfer (lower bars: the reverse). We note that examining positive transfer alone is insufficient to capture synergy, as gains on new samples may come at the cost of losing previously correct predictions. Figure 5b further summarizes the overall impact by computing the net difference between the upper and lower bars. **SyMerge** achieves the smallest net difference, demonstrating that it best preserves individual model knowledge while benefiting from merging.

Merging across disjoint basins. Our theoretical analysis assumes that models reside within a shared optimization basin (*i.e.*, cross-task linearity holds). To rigorously test **SyMerge** under conditions where this assumption is violated, we conducted a stress test by merging models derived from *different initializations*, effectively forcing the merge across disjoint loss basins where a significant loss barrier exists. We merge a DTD model fine-tuned from OpenAI-ViT (Radford et al., 2021) and a EuroSAT model from DataComp-ViT (Gadre et al., 2023). Since standard Task Arithmetic is inapplicable due to the lack of a common pre-trained weight, we defined task vectors relative to the simple weight average of the two models.

Table 5 illustrates a catastrophic failure for standard methods. Weight Averaging collapses (6.9%), confirming the presence of a severe loss barrier between the two models. Surgery also fails to recover performance (25.6%), as external adapters cannot rectify a fundamentally misaligned backbone. In contrast, **SyMerge** successfully bridges this gap, recovering the average accuracy to 79.1%. This result indicates that **SyMerge** is not merely applying distillation; it provides a practical solution for integrating models with severe functional or parameter-level discrepancies that cannot be resolved by existing merging methods. By jointly optimizing the merging coefficients and the internal layer, **SyMerge** effectively realigns the conflicting representations even when the parameter-space linearity breaks down.

Component analysis. We ablate our joint optimization strategy to confirm that both the merging coefficients and the

Table 6. Ablation on optimization components. Merging coefficients and task-specific layer are highly complementary.

Trainable		Avg. on N Tasks		
Coef	Layer	$N=8$	$N=14$	$N=20$
✓		84.6	82.4	81.6
	✓	88.2	83.0	75.0
✓	✓	90.1	88.7	88.6

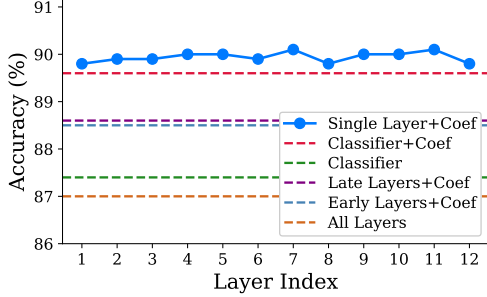
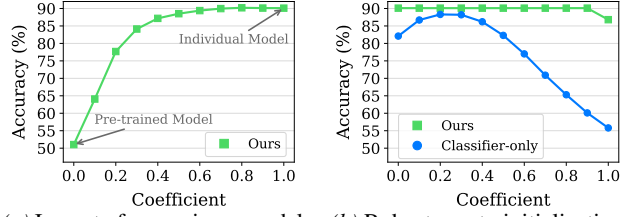


Figure 6. Impact of training different task-specific layers. We compare merged model performance when different CLIP ViT-B/32 layers are made trainable. The x-axis denotes the trainable layer index; single-layer adaptation remains stable across all 12 layers, whereas adapting multiple layers degrades performance.

task-specific layer are necessary for optimal performance. In Table 6, the results confirm that jointly optimizing both yields a significant synergistic gain, consistently outperforming the optimization of either component alone. The analysis reveals a key trade-off: layer-only adaptation excels on 8 tasks but scales poorly as task interference grows in the fixed shared encoder. Conversely, coefficient-only optimization is more robust against an increasing number of tasks but is ultimately limited by a sub-optimal, unaligned task-specific layer. Notably, the merging coefficients add fewer than 0.001M trainable parameters, yet they substantially improve the 20-task result over layer-only adaptation (75.0% $\rightarrow$ 88.6%). This suggests that the two components are highly complementary, and that the gain arises from their synergy rather than increased trainable capacity. **SyMerge**’s joint optimization refines the shared encoder while concurrently adapting the task-specific layer to its merged representations, creating the most robust and scalable strategy.

Effect of adjusting different layers. A natural question is whether the improvements observed in **SyMerge** are specific to classifier training. To explore this, we analyze the effect of training different layers beyond the classifier by updating merging coefficients along with each of the 12 transformer layers in the merged encoder. As shown in Figure 6, training a single internal layer achieves accuracy comparable to training the classifier, suggesting that task-specific information can be effectively captured by refining a single layer rather than modifying the entire model. Identifying the optimal layer could further enhance performance. On the other hand, training multiple layers at once, either



(a) Impact of supervisory model (b) Robustness to initialization

Figure 7. More analyses with merging coefficients. (a) Refined supervisory models merged under various coefficients are tested. Our design choice – using the unmerged individual models – performs near-optimally. (b) Our method shows strong robustness to different initial merging coefficients.

from the early stage (layers 1–6) or the later stage (layers 7–12), leads to a performance drop. This suggests that the gain is not merely due to increased trainable capacity, but comes from targeted single-layer adaptation that preserves task-agnostic knowledge. See Table E for details.

Improvement beyond classifier adaptation. One might suspect that our performance gains stem primarily from fine-tuning the task-specific layer rather than improving the merged encoder representations. To isolate the quality of the jointly optimized merged encoder, we conduct an ablation study using ViT-B/32 across 8 tasks where we retain the merged encoder learned by **SyMerge** but replace the fine-tuned classifier with the original zero-shot classifier. Remarkably, this configuration achieves an average accuracy of 82.4%, representing a substantial improvement over the standard Task Arithmetic baseline with the same zero-shot classifier (69.1%). This confirms that **SyMerge** significantly enhances the quality of the underlying merged encoder itself, independent of the classifier adaptation. Refer to Table D for per-task performance.

More studies with merging coefficients. We study whether individual model predictions are sufficient as supervisory signals. To refine them, we interpolate each individual model with the pre-trained model by scaling its task vector with a merging coefficient $\lambda \in [0, 1]$ (where $\lambda = 0$ and $\lambda = 1$ represent the pre-trained and individual models, respectively). Figure 7a shows that performance improves as λ increases, suggesting that guidance closer to the task-agnostic pre-trained model misses task-specific details. Interestingly, a combined model ($\lambda = 0.8$) performs best, which is about 90.2%. However, tuning λ would require additional hyperparameter search, so using the individual models directly is a practical option. Efficiently searching for the coefficient is a promising direction for future work. Also, our method benefits from robust merging coefficient initialization. Figure 7b shows that training only the classifier (*i.e.*, with a fixed shared encoder) is sensitive to initialization, whereas jointly training the classifier and coefficients substantially improves robustness.

Table 7. **Robustness to a corrupted expert.** In the 8-task ViT-B/32 setting, replacing one expert with a random-weight checkpoint severely degrades baselines, while **SyMerge** remains robust.

Method	Clean	With corrupted GTSRB expert		
	Avg. $\uparrow$	Avg. $\uparrow$	GTSRB $\uparrow$	Other-7 Avg. $\uparrow$
Task Arithmetic	69.1	6.0	4.8	6.2
AdaMerging	80.1	4.6	2.1	4.9
Surgery	87.5	18.8	1.7	21.2
WEMoE	89.4	4.7	5.4	4.6
SyMerge	90.1	77.3	1.7	88.1

Table 8. **Robustness to a backdoored expert.** With one expert backdoored, **SyMerge** achieves the best clean/backdoored accuracy and the lowest off-task attack success rate (ASR).

Method	Clean $\uparrow$	Backdoor $\uparrow$	Off-task ASR $\downarrow$
Task Arithmetic	76.3	76.0	95.6
Ties-Merging	74.5	74.4	90.6
AdaMerging	83.1	82.9	98.3
Surgery	84.5	84.6	86.6
SyMerge	86.1	86.0	75.6

Robustness to unreliable experts. To further examine the effect of unreliable expert supervision, we conduct two additional evaluations on ViT-B/32. First, we consider corrupted experts in the 8-task setting (Ilharco et al., 2023) by replacing the GTSRB expert with a Gaussian random-weight model matched in mean and variance. As shown in Table 7, **SyMerge** drops from 90.1 to 77.3 average accuracy, but the degradation is largely confined to the corrupted task, with the remaining tasks mostly preserved. In contrast, other adaptive baselines suffer collapse across tasks even when only one expert is corrupted. This suggests that **SyMerge** can limit the impact of a single corrupted expert, although it is not robust to arbitrary severe corruption.

Second, we evaluate backdoored experts following BadMerging (Zhang et al., 2024), where one of six experts, CIFAR100, is replaced with a model that behaves normally on clean inputs but produces attacker-specified outputs when a trigger patch is present. We report the off-task backdoor setting, which measures whether the trigger effect transfers from the backdoored expert to other tasks, providing a diagnostic test of cross-task vulnerability. As shown in Table 8, **SyMerge** achieves the highest clean accuracy (merging only clean experts) and backdoored accuracy (including one backdoored expert), while also obtaining the lowest off-task attack success rate among adaptive merging methods. These results indicate that **SyMerge** remains more stable than existing adaptive approaches under intentionally poisoned expert supervision.

Computational cost. Table 9 compares adaptation cost in the 8-task ViT-B/32 setting. **SyMerge** achieves 90.1% aver-

Table 9. **Efficiency comparison on the 8-task ViT-B/32 setting.** Peak GPU memory includes expert inference under the sequential-update setting. Runtime is measured per adaptation step. $\dagger$ denotes runtime with cached expert predictions.

Method	Params. (M)	GPU Mem. (GB)	Runtime (ms)
AdaMerging	< 0.001	7.09	114
Surgery	0.13	7.13	211
WEMoE	7.16	8.01	428
SyMerge	0.39	7.12	215 (111 $\dagger$)

age accuracy while requiring only 0.39M trainable parameters, compared to WEMoE with 89.4% accuracy and 7.16M trainable parameters. This suggests that **SyMerge** gains from task synergy rather than increased trainable capacity. Expert supervision is performed under a sequential per-task update strategy: at each step, only the expert for the current task is used to generate pseudo-labels, so there is no need to load all experts onto the GPU simultaneously. Including expert inference, **SyMerge** uses 7.12GB of peak memory and runs at 215ms per step; caching expert predictions reduces the runtime to 111ms, close to AdaMerging. Thus, **SyMerge** introduces manageable overhead while preserving lightweight adaptation. See Appendix E.6 for details.

5. Conclusion

In this work, we introduce **SyMerge**, an effective method that redefines the goal of model merging from mitigating interference to creating synergy, where tasks mutually enhance one another. **SyMerge** achieves this via a lightweight, test-time adaptive process: it jointly optimizes task-vector coefficients and a single task-specific layer on unlabeled data, using a robust self-labeling strategy for stable supervision. Our experiments confirm the broad effectiveness of **SyMerge**, which sets a new state-of-the-art across diverse benchmarks, including vision, dense prediction, and NLP, while scaling robustly as the number of tasks increases. Notably, **SyMerge** succeeds even in disjoint basins where prior baselines fail. Finally, the learned adaptation is transferable and intrinsically improves the merged encoder beyond task-specific tuning. Due to its strong performance and robustness in challenging regimes, **SyMerge** offers a practical solution for building powerful, scalable multi-task models.

Limitation. As with most model merging methods, **SyMerge** depends on the quality of individual experts, since it uses their predictions as self-labels. Therefore, it can still be affected by poor or biased experts despite its empirical robustness in our stress tests (Tables 7 and 8). **SyMerge** requires access to unlabeled target inputs for adaptation, which may be restrictive under privacy or latency constraints; further characterizing the interaction between coefficient tuning and single-layer adaptation remains future work.

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Impact Statement

This paper presents work whose goal is to advance the field of machine learning. There are many potential societal consequences of our work, none of which we feel must be specifically highlighted here.

References

- Caruana, R. Multitask learning. *Machine learning*, 28: 41–75, 1997.
- Cer, D., Diab, M., Agirre, E., Lopez-Gazpio, I., and Specia, L. SemEval-2017 task 1: Semantic textual similarity multilingual and crosslingual focused evaluation. In *Proc. of Int’l Workshop on Semantic Evaluation (SemEval)*, pp. 1–14, Vancouver, Canada, August 2017. Association for Computational Linguistics. doi: 10.18653/v1/S17-2001. URL <https://aclanthology.org/S17-2001/>.
- Chen, Z., Badrinarayanan, V., Lee, C.-Y., and Rabinovich, A. GradNorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proc. of Int’l Conf. on Machine Learning (ICML)*, pp. 794–803. PMLR, 2018.
- Choi, S., Zhang, Y., and Hong, S. Intra-inter modal attention blocks for rgb-d semantic segmentation. In *Proc. of Int’l Conf. on Multimedia Retrieval (ICMR)*, pp. 217–225, 2023.
- Chung, H. W., Hou, L., Longpre, S., Zoph, B., Tay, Y., Fedus, W., Li, Y., Wang, X., Dehghani, M., Brahma, S., et al. Scaling instruction-finetuned language models. *Journal of Machine Learning Research (JMLR)*, 25(70): 1–53, 2024.
- Csiszár, A., Körösi-Szabó, P., Matszangosz, A., Papp, G., and Varga, D. Similarity and matching of neural network representations. In *Proc. of Neural Information Processing Systems (NeurIPS)*, volume 34, pp. 5656–5668, 2021.
- Davari, M. and Belilovsky, E. Model breadcrumbs: Scaling multi-task model merging with sparse masks. In *Proc. of European Conf. on Computer Vision (ECCV)*, 2024.
- Deep, P. T., Bhardwaj, R., and Poria, S. Della-merging: Reducing interference in model merging through magnitude-based sampling. *arXiv preprint arXiv:2406.11617*, 2024.
- Deng, J., Dong, W., Socher, R., Li, L.-J., Li, K., and Fei-Fei, L. Imagenet: A large-scale hierarchical image database. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 248–255. Ieee, 2009.
- Dolan, B. and Brockett, C. Automatically constructing a corpus of sentential paraphrases. In *Proc. of Int’l Workshop on Paraphrasing (IWP)*, 2005.
- Du, G., Lee, J., Li, J., Jiang, R., Guo, Y., Yu, S., Liu, H., Goh, S. K., Tang, H.-K., He, D., et al. Parameter competition balancing for model merging. *Proc. of Neural Information Processing Systems (NeurIPS)*, 37:84746–84776, 2024.
- Gadre, S. Y., Ilharco, G., Fang, A., Hayase, J., Smyrnis, G., Nguyen, T., Marten, R., Wortsman, M., Ghosh, D., Zhang, J., et al. Datacomp: In search of the next generation of multimodal datasets. In *Proc. of Neural Information Processing Systems (NeurIPS)*, 2023.
- Gargiulo, A. A., Crisostomi, D., Bucarelli, M. S., Scardapane, S., Silvestri, F., and Rodola, E. Task singular vectors: Reducing task interference in model merging. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, 2025.
- Giampiccolo, D., Magnini, B., Dagan, I., and Dolan, W. B. The third pascal recognizing textual entailment challenge. In *Proc. of ACL-PASCAL Workshop on Textual Entailment and Paraphrasing*, pp. 1–9, 2007.
- He, K., Zhang, X., Ren, S., and Sun, J. Deep residual learning for image recognition. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778, 2016.
- Hendrycks, D. and Dietterich, T. Benchmarking neural network robustness to common corruptions and perturbations. In *Proc. of Int’l Conf. on Learning Representations (ICLR)*, 2019.
- Hu, Y., Xian, R., Wu, Q., Fan, Q., Yin, L., and Zhao, H. Revisiting scalarization in multi-task learning: A theoretical perspective. In *Proc. of Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- Huang, C., Ye, P., Chen, T., He, T., Yue, X., and Ouyang, W. Emr-merging: Tuning-free high-performance model merging. *Proc. of Neural Information Processing Systems (NeurIPS)*, 2024a.

- Huang, H., Huang, Y., Lin, L., Tong, R., Chen, Y.-W., Zheng, H., Li, Y., and Zheng, Y. Going beyond multi-task dense prediction with synergy embedding models. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 28181–28190, 2024b.
- Ilharcó, G., Wortsman, M., Gadre, S. Y., Song, S., Hajishirzi, H., Kornblith, S., Farhadi, A., and Schmidt, L. Patching open-vocabulary models by interpolating weights. *Proc. of Neural Information Processing Systems (NeurIPS)*, 35: 29262–29277, 2022.
- Ilharcó, G., Ribeiro, M. T., Wortsman, M., Gururangan, S., Schmidt, L., Hajishirzi, H., and Farhadi, A. Editing models with task arithmetic. In *Proc. of Int’l Conf. on Learning Representations (ICLR)*, 2023.
- Iyer, S., Dandekar, N., and Csernai, K. First quora dataset release: Question pairs, 2017. URL <https://data.quora.com/First-Quora-Dataset-Release-Question-Pairs>.
- Jeong, W. and Yoon, K.-J. Selective task group updates for multi-task optimization. In *Proc. of Int’l Conf. on Learning Representations (ICLR)*, 2025.
- Jin, R., Hou, B., Xiao, J., Su, W., and Shen, L. Fine-tuning attention modules only: Enhancing weight disentanglement in task arithmetic. In *Proc. of Int’l Conf. on Learning Representations (ICLR)*, 2025.
- Kendall, A., Gal, Y., and Cipolla, R. Multi-task learning using uncertainty to weigh losses for scene geometry and semantics. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 7482–7491, 2018.
- Kim, Y., Lee, S., Jung, A., Ryu, B., and Hong, S. Task vector quantization for memory-efficient model merging. In *Proc. of Int’l Conf. on Computer Vision (ICCV)*, pp. 20105–20115, 2025.
- Kingma, D. P. Adam: A method for stochastic optimization. *Proc. of Int’l Conf. on Learning Representations (ICLR)*, 2015.
- Kirkpatrick, J., Pascanu, R., Rabinowitz, N., Veness, J., Desjardins, G., Rusu, A. A., Milan, K., Quan, J., Ramalho, T., Grabska-Barwinska, A., et al. Overcoming catastrophic forgetting in neural networks. *Proceedings of the national academy of sciences*, 114(13):3521–3526, 2017.
- Lee, D.-H. et al. Pseudo-label: The simple and efficient semi-supervised learning method for deep neural networks. In *Proc. of Int’l Conf. on Machine Learning Workshops (ICMLW)*, volume 3, pp. 896. Atlanta, 2013.
- Li, B., Shen, C., Dai, Y., Van Den Hengel, A., and He, M. Depth and surface normal estimation from monocular images using regression on deep features and hierarchical crfs. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 1119–1127, 2015.
- Li, D., Sharma, A., and Zhang, H. R. Scalable multitask learning using gradient-based estimation of task affinity. In *Proc. of Int’l Conf. on Knowledge Discovery and Data Mining (KDD)*, pp. 1542–1553, 2024.
- Liang, J., He, R., and Tan, T. A comprehensive survey on test-time adaptation under distribution shifts. *Int’l Journal of Computer Vision (IJCV)*, 133(1):31–64, 2025.
- Lin, S., Yang, L., Fan, D., and Zhang, J. Beyond not-forgetting: Continual learning with backward knowledge transfer. In *Proc. of Neural Information Processing Systems (NeurIPS)*, volume 35, pp. 16165–16177, 2022.
- Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., and Stoyanov, V. Roberta: A robustly optimized bert pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- Lopez-Paz, D. and Ranzato, M. Gradient episodic memory for continual learning. In *Proc. of Neural Information Processing Systems (NeurIPS)*, volume 30, 2017.
- Marczak, D., Twardowski, B., Trzciński, T., and Cygert, S. Magmax: Leveraging model merging for seamless continual learning. In *Proc. of European Conf. on Computer Vision (ECCV)*, pp. 379–395. Springer, 2024.
- Marczak, D., Magistri, S., Cygert, S., Twardowski, B., Bagdanov, A. D., and van de Weijer, J. No task left behind: Isotropic model merging with common and task-specific subspaces. In *Proc. of Int’l Conf. on Machine Learning (ICML)*, 2025.
- Min, J., Jeon, Y., Kim, J., and Choi, M. S2m2: Scalable stereo matching model for reliable depth estimation. In *Proc. of Int’l Conf. on Computer Vision (ICCV)*, pp. 26729–26739, 2025.
- Misra, I., Shrivastava, A., Gupta, A., and Hebert, M. Cross-stitch networks for multi-task learning. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 3994–4003, 2016.
- Netzer, Y., Wang, T., Coates, A., Bissacco, A., Wu, B., Ng, A. Y., et al. Reading digits in natural images with unsupervised feature learning. In *Proc. of Neural Information Processing Systems Workshops (NeurIPSW)*, volume 2011, pp. 4. Granada, 2011.
- Oh, C., Lim, H., Kim, M., Han, D., Yun, S., Choo, J., Hauptmann, A., Cheng, Z.-Q., and Song, K. Towards calibrated robust fine-tuning of vision-language models. In *Proc. of Neural Information Processing Systems (NeurIPS)*, 2024.

- Oh, C., Li, Y., Song, K., Yun, S., and Han, D. Dawin: Training-free dynamic weight interpolation for robust adaptation. In *Proc. of Int'l Conf. on Learning Representations (ICLR)*, 2025.
- Ortiz-Jimenez, G., Favero, A., and Frossard, P. Task arithmetic in the tangent space: Improved editing of pre-trained models. *Proc. of Neural Information Processing Systems (NeurIPS)*, 36:66727–66754, 2023.
- Panariello, A., Marczak, D., Magistri, S., Porrello, A., Twardowski, B., Bagdanov, A. D., Calderara, S., and van de Weijer, J. Accurate and efficient low-rank model merging in core space. In *Proc. of Neural Information Processing Systems (NeurIPS)*, 2025.
- Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *Proc. of Int'l Conf. on Machine Learning (ICML)*, pp. 8748–8763. PMLR, 2021.
- Rajpurkar, P., Zhang, J., Lopyrev, K., and Liang, P. SQuAD: 100,000+ questions for machine comprehension of text. In Su, J., Duh, K., and Carreras, X. (eds.), *Proc. of Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 2383–2392, Austin, Texas, November 2016. Association for Computational Linguistics. doi: 10.18653/v1/D16-1264. URL <https://aclanthology.org/D16-1264/>.
- Riemer, M., Cases, I., Ajemian, R., Liu, M., Rish, I., Tu, Y., and Tesauro, G. Learning to learn without forgetting by maximizing transfer and minimizing interference. In *Proc. of Int'l Conf. on Learning Representations (ICLR)*, 2019.
- Silberman, N., Hoiem, D., Kohli, P., and Fergus, R. Indoor segmentation and support inference from rgb-d images. In *Proc. of European Conf. on Computer Vision (ECCV)*, pp. 746–760. Springer, 2012.
- Socher, R., Perelygin, A., Wu, J., Chuang, J., Manning, C. D., Ng, A. Y., and Potts, C. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proc. of Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, pp. 1631–1642, 2013.
- Stallkamp, J., Schlipsing, M., Salmen, J., and Igel, C. The german traffic sign recognition benchmark: a multi-class classification competition. In *Proc. of International Joint Conference on Neural Networks*, pp. 1453–1460. IEEE, 2011.
- Standley, T., Zamir, A., Chen, D., Guibas, L., Malik, J., and Savarese, S. Which tasks should be learned together in multi-task learning? In *Proc. of Int'l Conf. on Machine Learning (ICML)*, pp. 9120–9132. PMLR, 2020.
- Sun, W., Li, Q., Wang, W., Liu, Y., Geng, Y.-a., and Li, B. Towards minimizing feature drift in model merging: Layer-wise task vector fusion for adaptive knowledge integration. In *Proc. of Neural Information Processing Systems (NeurIPS)*, 2025.
- Tang, A., Shen, L., Luo, Y., Yin, N., Zhang, L., and Tao, D. Merging multi-task models via weight-ensembling mixture of experts. In *Proc. of Int'l Conf. on Machine Learning (ICML)*, 2024a.
- Tang, A., Shen, L., Luo, Y., Zhan, Y., Hu, H., Du, B., Chen, Y., and Tao, D. Parameter efficient multi-task model fusion with partial linearization. In *Proc. of Int'l Conf. on Learning Representations (ICLR)*, 2024b.
- Tang, A., Shen, L., Luo, Y., Yang, E., Hu, H., Zhang, L., Du, B., and Tao, D. Fusionbench: A unified library and comprehensive benchmark for deep model fusion. *Journal of Machine Learning Research (JMLR)*, 26(307):1–38, 2025. URL <http://jmlr.org/papers/v26/25-1243.html>.
- Vandenhende, S., Georgoulis, S., Van Gansbeke, W., Proesmans, M., Dai, D., and Van Gool, L. Multi-task learning for dense prediction tasks: A survey. *IEEE Trans. on Pattern Anal. Mach. Intell. (TPAMI)*, 44(7):3614–3633, 2021.
- Wang, A., Singh, A., Michael, J., Hill, F., Levy, O., and Bowman, S. R. Glue: A multi-task benchmark and analysis platform for natural language understanding. In *Proc. of Int'l Conf. on Learning Representations (ICLR)*, 2019.
- Wang, K., Dimitriadis, N., Ortiz-Jiménez, G., Fleuret, F., and Frossard, P. Localizing task information for improved model merging and compression. In *Proc. of Int'l Conf. on Machine Learning (ICML)*, 2024.
- Wang, K., Dimitriadis, N., Favero, A., Ortiz-Jimenez, G., Fleuret, F., and Frossard, P. Lines: Post-training layer scaling prevents forgetting and enhances model merging. In *Proc. of Int'l Conf. on Learning Representations (ICLR)*, 2025.
- Warstadt, A., Singh, A., and Bowman, S. R. Neural network acceptability judgments. *Transactions of the Association for Computational Linguistics (TACL)*, 7:625–641, 2019.
- Wei, Q., He, S., Yang, E., Liu, T., Wang, H., Feng, L., and An, B. Representation surgery in model merging with probabilistic modeling. In *Proc. of Int'l Conf. on Machine Learning (ICML)*, 2025.
- Williams, A., Nangia, N., and Bowman, S. A broad-coverage challenge corpus for sentence understanding through inference. In *Proc. of the Conf. of the North American Chapter of the Association for Computational*

- Linguistics (NAACL)*, pp. 1112–1122, New Orleans, Louisiana, June 2018. Association for Computational Linguistics. doi: 10.18653/v1/N18-1101. URL <https://aclanthology.org/N18-1101/>.
- Wortsman, M., Ilharco, G., Gadre, S. Y., Roelofs, R., Gontijo-Lopes, R., Morcos, A. S., Namkoong, H., Farhadi, A., Carmon, Y., Kornblith, S., et al. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *Proc. of Int’l Conf. on Machine Learning (ICML)*, pp. 23965–23998. PMLR, 2022a.
- Wortsman, M., Ilharco, G., Kim, J. W., Li, M., Kornblith, S., Roelofs, R., Lopes, R. G., Hajishirzi, H., Farhadi, A., Namkoong, H., et al. Robust fine-tuning of zero-shot models. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 7959–7971, 2022b.
- Xu, J., Li, J., and Zhang, J. Scalable model merging with progressive layer-wise distillation. In *Proc. of Int’l Conf. on Machine Learning (ICML)*, 2025.
- Yadav, P., Tam, D., Choshen, L., Raffel, C. A., and Bansal, M. Ties-merging: Resolving interference when merging models. In *Proc. of Neural Information Processing Systems (NeurIPS)*, volume 36, 2023.
- Yang, E., Shen, L., Wang, Z., Guo, G., Chen, X., Wang, X., and Tao, D. Representation surgery for multi-task model merging. In *Proc. of Int’l Conf. on Machine Learning (ICML)*, 2024a.
- Yang, E., Wang, Z., Shen, L., Liu, S., Guo, G., Wang, X., and Tao, D. Adamerging: Adaptive model merging for multi-task learning. In *Proc. of Int’l Conf. on Learning Representations (ICLR)*, 2024b.
- Yu, L., Yu, B., Yu, H., Huang, F., and Li, Y. Language models are super mario: Absorbing abilities from homologous models as a free lunch. In *Proc. of Int’l Conf. on Machine Learning (ICML)*, 2024.
- Yu, T., Kumar, S., Gupta, A., Levine, S., Hausman, K., and Finn, C. Gradient surgery for multi-task learning. *Proc. of Neural Information Processing Systems (NeurIPS)*, 33: 5824–5836, 2020.
- Zamir, A. R., Sax, A., Shen, W., Guibas, L. J., Malik, J., and Savarese, S. Taskonomy: Disentangling task transfer learning. In *Proc. of Computer Vision and Pattern Recognition (CVPR)*, pp. 3712–3722, 2018.
- Zhang, J., Chi, J., Li, Z., Cai, K., Zhang, Y., and Tian, Y. Badmerging: Backdoor attacks against model merging. In *Proc. of ACM SIGSAC Conf. on Computer and Communications Security (CCS)*, pp. 4450–4464, 2024.
- Zhang, Y., Choi, S., and Hong, S. Spatio-channel attention blocks for cross-modal crowd counting. In *Proc. of Asian Conf. on Computer Vision (ACCV)*, pp. 90–107, 2022.
- Zhang, Y., Choi, S., and Hong, S. Memory-efficient cross-modal attention for rgb-x segmentation and crowd counting. *Pattern Recognition (PR)*, 162:111376, 2025a.
- Zhang, Y., Kim, Y., Choi, Y.-G., Kim, H., Liu, H., and Hong, S. Backpropagation-free test-time adaptation via probabilistic gaussian alignment. *Proc. of Neural Information Processing Systems (NeurIPS)*, 38:76280–76310, 2025b.
- Zhang, Y., Liu, H., Kim, Y., and Hong, S. CAT-TPT: Class-agnostic text-based test-time prompt tuning for vision-language models. *Int’l Journal of Computer Vision (IJCV)*, 133(10):6930–6952, 2025c.
- Zheng, S. and Wang, H. Free-merging: Fourier transform for efficient model merging. In *Proc. of Int’l Conf. on Computer Vision (ICCV)*, 2025.
- Zhou, Z., Chen, Z., Chen, Y., Zhang, B., and Yan, J. On the emergence of cross-task linearity in the pretraining-finetuning paradigm. In *Proc. of Int’l Conf. on Machine Learning (ICML)*, 2024.

Appendix

This appendix provides an overview of our experimental setup and baselines, further experimental results across tasks, a theoretical proof, and additional empirical analyses. The detailed descriptions of each section are summarized as follows:

- Appendix A: Related work;
- Appendix B: Detailed experimental setup and hyperparameter configurations;
- Appendix C: Additional results for motivation, pilot study, vision, and NLP tasks;
- Appendix D: Proof for Proposition 3.1;
- Appendix E: Further analyses on task-specific layers, loss functions, convergence, cross-task linearity, confidence-based filtering, computational efficiency, and sparsity.

A. Related Work

A.1. Task Interference in Multi-task Model Merging

To overcome the computational cost and task conflicts (Chen et al., 2018; Yu et al., 2020) of traditional Multi-Task Learning (MTL) (Caruana, 1997; Vandenhende et al., 2021), model merging has emerged as an efficient alternative that combines independently fine-tuned models without expensive joint training (Wortsman et al., 2022a;b; Ilharco et al., 2022).

A foundational work in model merging for MTL is Task Arithmetic (Ilharco et al., 2023), which introduces task vectors, defined as the difference between pre-trained and fine-tuned models. This concept enables the arithmetic combination of multiple models. Subsequent research has focused on refining this merging process. Yadav et al. (2023) and Du et al. (2024) directly address interference by assessing intra-task parameter significance and inter-task conflicts, and then removing parameters identified as redundant or conflicting. Departing from using flattened task vectors, Gargiulo et al. (2025) and Marczak et al. (2025) leverage low-rank approximation to analyze the structural information preserved in per-layer task matrices, thereby resolving conflicts between tasks. Test-time adaptive approaches improve merging by either adjusting merging coefficients to mitigate interference (Yang et al., 2024b; Tang et al., 2024a) or tuning additional modules to minimize feature discrepancy between individual and merged models (Yang et al., 2024a; Wei et al., 2025).

A distinct line of research seeks to minimize task interference before merging by pursuing weight disentanglement, an ideal where task vectors do not affect each other at all (Ortiz-Jimenez et al., 2023; Jin et al., 2025; Tang et al., 2024b). We argue the conventional goal of non-interference is too limited. We propose the more ambitious objective of creating synergy, where tasks actively improve each other’s performance through cross-task enhancement, rather than merely avoiding conflict.

A.2. Task Synergy and Positive Transfer

Classical MTL views learning from related tasks as a source of inductive bias that can improve generalization (Caruana, 1997). Later studies analyze task relationships, task affinities, and task groupings, including distinctions between cooperative and competing tasks (Zamir et al., 2018; Standley et al., 2020). Recent work further studies task interaction through task affinity estimation and adaptive grouping methods, aiming to reduce negative transfer by identifying or selecting cooperative task groups (Li et al., 2024; Jeong & Yoon, 2025). Huang et al. (2024b) model cross-task interaction more explicitly through synergy embeddings.

Continual learning makes a similar distinction between interference and transfer. Forward transfer measures whether previously acquired knowledge improves future tasks, while backward transfer measures whether learning a new task improves previous tasks (Lopez-Paz & Ranzato, 2017; Riemer et al., 2019; Lin et al., 2022). These works motivate a broader view in which learning systems should not merely avoid forgetting, conflict, or interference, but should exploit beneficial interactions across tasks.

SyMerge differs in that it studies positive transfer in post-hoc model merging. Independently fine-tuned experts are merged after training, and synergy manifests as cross-task functional compatibility between the merged encoder and task-specific predictors. Accordingly, we evaluate synergy not only by average accuracy but also by sample-level positive and negative transfer relative to each task expert.

B. Experimental Setup

B.1. Training Details

Main experiments. We initialize the layer-wise merging coefficients for all layers to 0.3 by default, following the values used in the previous method (Yang et al., 2024b). However, for the 14 and 20 vision tasks, the coefficients are initially set to 0.1, as using 0.3 incurred significantly lower Task Arithmetic (Ilharco et al., 2023) performance for these tasks, presumably due to the increased number of tasks. We use the Adam (Kingma, 2015) optimizer with momentum parameters (0.9, 0.999) to update the coefficients and the classifier. For vision tasks, the learning rate is set to 0.01 when training the classifier and 0.001 when training only the merging coefficients. For dense prediction tasks, the learning rate of 0.0001 is used. For NLP tasks, the learning rate of 0.0005 is used, regardless of whether the classifier, the merging coefficients, or both are being updated. Consistent with prior works (Yang et al., 2024b; Tang et al., 2024a; Yang et al., 2024a), vision tasks are trained over 500 iterations, using a batch size of 32. Dense prediction tasks are trained for 25 iterations with a batch size of 16. NLP tasks are trained over 1,000 iterations with a batch size of 32. For each task, the loss is computed, and the model parameters are updated immediately following the forward pass for each batch. To enable sequential processing, the order of tasks is randomized. To ensure a fair evaluation, the number of samples per task and all other experimental settings are identically applied to the other test-time adaptive model merging methods.

Discussions on implementation. Previous methods (Yang et al., 2024b; Tang et al., 2024a) sample a batch for each task, pass each batch through a shared backbone, and then process it through the corresponding task-specific head, which is typical. The losses from each task are then combined, and a single gradient update is performed after completing the forward pass for all tasks. However, this approach becomes less efficient as the number of tasks or model size increases.

To address this, we experiment with a sequential update strategy, where updates are applied immediately after each task’s forward pass instead of the typical one. Interestingly, this sequential update approach gives a positive byproduct: performance improvements over the traditional approach. While the original AdaMerging achieves an average accuracy of 80.1%, our sequential update strategy improves it to 81.6% for the 8 vision tasks using ViT-B/32. Similarly, our method also enjoys a gain when using sequential updates. We argue that this improvement may result from mitigating catastrophic forgetting in continual learning (Kirkpatrick et al., 2017) by training tasks sequentially rather than simultaneously.

For vision tasks, we employ a confidence-based filtering strategy to ensure the model learns from reliable supervisory signals. Motivated by uncertainty-based reliability cues used in robust adaptation (Oh et al., 2025) and the need for calibrated confidence for reliable predictions (Oh et al., 2024), we use a relative confidence-based criterion instead of a fixed threshold. Specifically, we exclude samples where the merged model’s top-1 confidence is higher than that of the individual models. This ensures the merged model is supervised by more reliable data, leading to more effective adaptation and improved performance.

Pilot study. The aforementioned training details are applied almost identically to the experiments in the pilot study of our main paper. As stated in the main paper, only the classifier was chosen (as a straightforward and practical option) to train on the given Task Arithmetic merged weights for the 8 vision tasks using ViT-B/32. A difference is that, instead of using the test dataset, the training dataset is used in a supervised manner to produce the results closer to the upper bound, in which the difference stands out more. We train the models for only one epoch.

B.2. Baseline Details

We compare our method against a diverse set of baselines, ranging from simple merging strategies to advanced methods that employ different mechanisms to address conflicts between tasks.

Pretrained indicates a model that predicts multiple tasks without additional fine-tuning for task-specific requirements. However, the absence of task-specific information for downstream tasks generally leads to poor performance.

Individual refers to the fine-tuning of individual pre-trained models for each task. Since there is no interference between tasks, it has commonly been used as a strong reference for task-specific performance.

Weight Averaging merges multiple individual models by directly averaging their parameters to create a single model for multi-task learning. Although simple, this method lacks task-specific adjustments.

Task Arithmetic (Ilharco et al., 2023) defines the difference between the fine-tuned and pre-trained model parameters as a task vector. By combining multiple task vectors and adding them to the pre-trained model, it enables multi-task learning.

Table A. Performance comparison on the original test set versus the average of corrupted test sets.

Method	Original Test Set					Corrupted Test Set (Average)				
	Cars	EuroSAT	RESISC45	GTSRB	Avg.	Cars	EuroSAT	RESISC45	GTSRB	Avg.
Individual	77.7	99.9	96.1	98.7	93.1	73.0	54.1	84.9	86.2	74.5
Task Arithmetic	67.0	94.0	82.6	75.1	79.7	61.3	46.6	68.6	47.0	55.9
Ties-Merging	67.5	83.7	79.3	65.4	74.0	62.6	41.3	67.1	39.1	52.5
PCB-Merging	70.8	89.6	84.6	80.7	81.4	65.3	46.0	72.1	52.9	59.1
LiNeS	69.5	96.3	84.6	82.4	83.2	64.1	50.1	70.4	53.1	59.4
Consensus TA	67.1	95.4	78.0	84.9	81.4	61.4	50.1	48.0	70.2	57.4
Iso-Merging	73.6	96.3	91.3	93.4	88.6	67.5	51.4	64.2	76.9	65.0
AdaMerging	76.2	96.0	87.5	97.0	89.2	68.5	44.4	74.6	81.4	67.2
Surgery w/ Ada	73.4	98.5	91.2	97.8	90.2	67.2	52.4	77.8	78.0	68.8
WEMoE	79.1	99.3	95.5	99.1	93.2	74.3	43.3	86.9	70.1	68.6
SyMerge	77.6	99.4	95.3	98.3	92.7	72.9	53.1	84.3	84.6	73.7

MagMax (Marczak et al., 2024) merges task vectors (Ilharco et al., 2023) by selecting the parameter with the largest magnitude for each position, consolidating knowledge into a single model without retaining task-specific data.

Ties Merging (Yadav et al., 2023) highlights the importance of addressing interference in task arithmetic-based merging. It involves removing redundant parameters from the task vector and resolving parameter sign conflicts.

PCB-Merging (Du et al., 2024) resolves conflicting parameter values that arise during model merging by using intra-task importance and inter-task similarity to rescale and prune task vectors.

LiNeS (Wang et al., 2025) observes that reducing the influence of shallow layers helps prevent distortion of general representations, thereby simplifying merging coefficient selection by allowing them to increase linearly with layer depth.

Consensus TA (Wang et al., 2024) enhances Task Arithmetic (Ilharco et al., 2023) by using Consensus Merging, which retains only weights beneficial to multiple tasks while eliminating irrelevant or task-specific weights. This process uses task-specific binary masks to identify relevant weights and forms a consensus mask to minimize task interference.

TSV-M (Gargiulo et al., 2025) decomposes per-layer task matrices using Singular Value Decomposition (SVD) to obtain low-rank Task Singular Vectors (TSVs). It then decorrelates them to mitigate interference when merging models.

ISO-Merging (Marczak et al., 2025) improves alignment by flattening the singular value spectrum of the merged task matrix, creating a uniform common subspace. To better preserve unique features, it further enhances this common subspace by incorporating task-specific singular vectors.

EMR-Merging (Huang et al., 2024a) involves three steps: Elect, Mask, and Rescale-Merging. These steps select key parameters to form a unified model, apply task-specific masks for each task, and adjust scales to achieve better performance.

AdaMerging (Yang et al., 2024b) adaptively learns merging coefficients at the task or layer level by minimizing the entropy of predictions on unlabeled test data.

Representation Surgery (Yang et al., 2024a) reduces representation bias by training a task-specific module that aligns the merged model’s features with those of the individual models.

WEMoE (Tang et al., 2024a) utilizes a Mixture of Experts (MoE) module to dynamically separate and integrate shared and task-specific knowledge based on input samples. By training the router on unlabeled test data, it optimizes routing weights and improves task-specific performance.

ProbSurgery (Wei et al., 2025) mitigates representation bias by modeling the bias as a learnable distribution to better capture the uncertainty that arises from parameter interference.

LOT Merging (Sun et al., 2025) mitigates feature drift through a convex quadratic optimization approach, which allows for efficient closed-form solutions requiring minimal data.

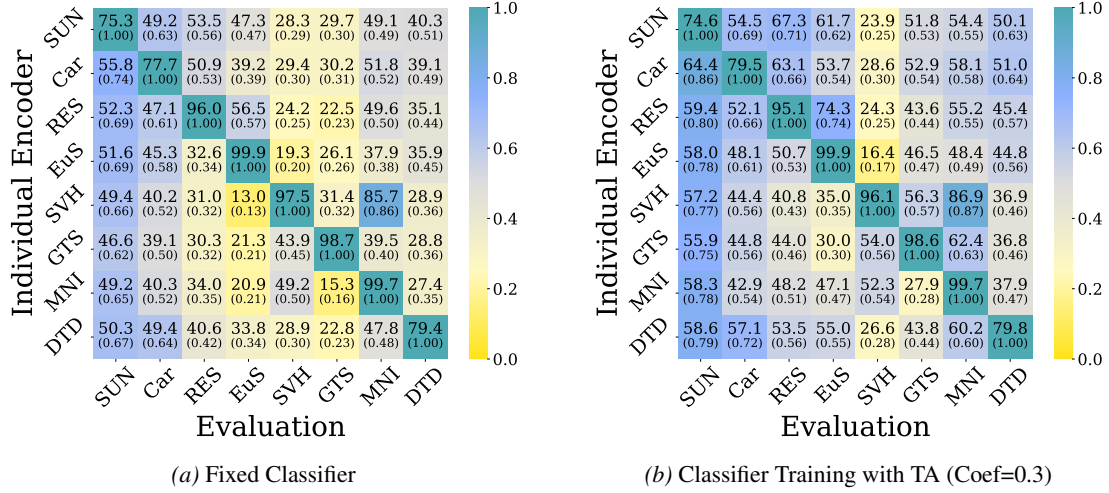


Figure A. Cross-task evaluations across diverse encoders and heads from different tasks. Each cell (i, j) displays the accuracy when features from task i 's visual encoder are classified by task j 's classifier. The upper value in a cell shows the absolute task accuracy, and the number below (in parentheses) denotes the relative performance compared to the diagonal. (a) Base (untrained) classifiers generally deteriorate when evaluated on other tasks or using different encoders. (b) Classifiers trained with Task Arithmetic (TA) features show mostly reduced performance loss (*i.e.*, higher relative numbers). This suggests that adjusting task-specific layers effectively achieves better functional alignment.

C. More Experimental Results

C.1. Experiments on corrupted test datasets

To evaluate the robustness of merging methods against distribution shifts, we conduct experiments on corrupted datasets. Following the protocol of ImageNet-C (Hendrycks & Dietterich, 2019), we apply 7 distinct types of corruption (*e.g.*, motion blur, impulse noise, gaussian noise, pixelate, spatter, contrast, and JPEG compression) at severity level 5 to the test sets of 4 vision tasks (Cars, EuroSAT, RESISC45, GTSRB). This setup provides a rigorous testbed for evaluating performance on out-of-distribution data. In the main paper, we presented the average performance across various corruption types and tasks to demonstrate the overall vulnerability of training-free methods to distribution shifts (Figure 1). This section provides a comprehensive breakdown of those results.

Table A shows the average performance across all 7 corruption types, revealing specific vulnerabilities in baseline methods. For instance, WEMoE and AdaMerging exhibit a significant performance drop on the EuroSAT task, while some training-free methods are particularly susceptible to GTSRB. In contrast, our method maintains significantly higher accuracy across all tasks, demonstrating superior robustness. For a more granular analysis, Table K details the performance of each merging method on all 4 tasks, individually evaluated across each of the 7 corruption types.

C.2. Pilot Study

Figure Aa shows cross-task evaluations via individual visual encoders with their respective base classifiers; Figure Ab shows results from combining individual visual encoders with trained classifiers from Task Arithmetic. We observe consistent improvements in cross-task performance when using the classifiers trained on other tasks. For instance, integrating the GTSRB (Stallkamp et al., 2011) classifier trained on Task Arithmetic features with the SVHN (Netzer et al., 2011) encoder achieves 56.3% accuracy (Figure Ab), surpassing the base classifier's 31.4% (Figure Aa). These results suggest that training the classifier on merged features for the GTSRB task also facilitates the alignment of individual SVHN features with the GTSRB task. This demonstrates the trained classifier's enhanced ability to achieve functional alignment with the encoder. Figure I presents the cross-task evaluation results of a classifier trained on a Task-Arithmetic merged encoder with merging coefficients ranging from 0.1 to 1.0, showing the task-pair results corresponding to the main paper's Figure 3b. From this, we can observe a clear variance in performance depending on the choice of the merging coefficient. This suggests that even when task-specific layers are trained, the choice of a shared encoder can lead to suboptimal performance.

Table B. Multi-task performance across 8 generative tasks. We present a performance comparison of merging the FLAN-T5 models fine-tuned on 8 NLP tasks in GLUE.

Method	CoLA	SST2	MRPC	STS-B	QQP	MNLI	QNLI	RTE	Avg.
Individual	75.0	83.4	87.5	91.5	85.4	85.9	93.6	88.7	86.4
Weight Average	69.1	91.7	79.4	73.2	83.9	62.6	89.8	81.2	78.9
Task Arithmetic (Ilharco et al., 2023)	70.5	92.3	78.4	77.8	83.6	57.8	90.2	80.5	78.9
DARE (Yu et al., 2024)	69.2	91.7	79.7	74.1	83.9	63.1	89.8	81.2	79.1
TIES-Merging (Yadav et al., 2023)	70.3	91.7	78.9	78.3	83.5	65.0	90.2	81.6	79.9
DARE + TIES	70.7	91.7	78.2	80.3	83.6	56.8	90.2	80.1	79.0
AdaMerging (Yang et al., 2024b)	70.9	90.7	78.7	67.4	81.1	80.4	89.4	82.0	80.1
WEMoE (Tang et al., 2024a)	70.7	91.5	78.2	72.5	82.4	76.8	90.0	78.7	80.1
SyMerge (Ours)	69.1	93.4	79.7	82.6	83.4	82.9	88.1	80.9	82.5

C.3. Vision Tasks

We reported the average performance of ViT-B-32 and ViT-L-14 on 8, 14, and 20 tasks in Table 1 of the main paper. In Tables L, M, N, O, P, and Q, we present the per-task performance of ViT-B-32 and ViT-L-14 for the 8-, 14-, and 20-task settings. For each task, our method consistently outperforms other approaches and achieves performance close to that of individual models.

C.4. NLP Tasks

We conduct experiments to investigate how the training of the classifier and merging coefficients, respectively, affects the performance of NLP tasks in **SyMerge**, as presented in Table 3 of the main paper. Figure B shows the results of merging RoBERTa (Liu et al., 2019) models for 8 NLP tasks using the predictions of individual models as guidance, optimized through the cross-entropy loss. As mentioned in the main paper, evaluation metrics differ across tasks: the Matthews correlation coefficient is used for CoLA (Warstadt et al., 2019), the average of Pearson and Spearman correlations is applied to STS-B (Cer et al., 2017), and the accuracy is used for all other tasks (Socher et al., 2013; Dolan & Brockett, 2005; Iyer et al., 2017; Williams et al., 2018; Rajpurkar et al., 2016; Giampiccolo et al., 2007). The approach where both the classifier and coefficients are trained simultaneously corresponds to **SyMerge**. To examine the role of classifier training in NLP tasks, ablation experiments are conducted by removing specific components. The results show that training only the coefficients leads to the lowest performance while training only the classifier achieves a relatively high performance. Furthermore, training both the classifier and coefficients together demonstrated a complementary effect, achieving the highest performance.

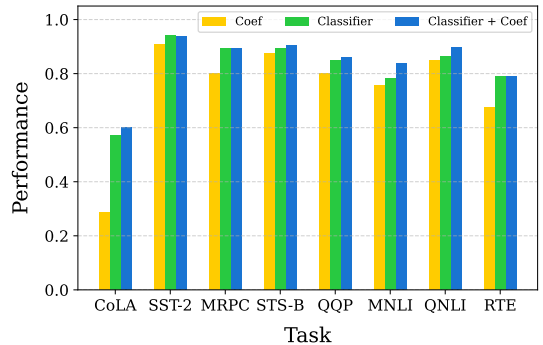


Figure B. Multi-task performance when merging RoBERTa models on 8 tasks. Yellow indicates training only coefficients, green indicates training only the classifier, and blue indicates training both in our method.

Additionally, we evaluate SyMerge on a larger sequence-to-sequence model, FLAN-T5 Base (Chung et al., 2024), using 8 GLUE tasks cast in a text-generation format to test scalability. We use a batch size of 4 for 1000 iterations, following the FusionBench protocol (Tang et al., 2025). As shown in Table B, SyMerge consistently outperforms test-time merging methods such as AdaMerging and WEMoE in this larger LM setting and reduces the gap to individually trained experts. Since generation models output token-level distributions, SyMerge applies directly by matching the merged model’s distribution to each expert’s distribution on unlabeled target text with cross-entropy.

C.5. Robustness to Data Scarcity

In real-world deployments, models often face a stream of queries or strictly limited data access rather than a large, pre-collected batch. To evaluate **SyMerge** in such data-scarce and practical scenarios, we conduct experiments in two distinct settings: *Online Test-Time Adaptation* and *Limited-Data Adaptation*.

Table C. **Robustness to Data Scarcity.** We evaluate **SyMerge** in (a) Online TTA (streaming) and (b) Limited-Data scenarios on ViT-B/32.

(a) Online TTA Performance.							(b) Data Efficiency.					
Method	Seen Samples						Method	Unlabeled Data Ratio				
	10	20	50	100	200	500		1%	5%	10%	20%	100%
AdaMerging	70.0	71.5	71.1	72.0	72.6	74.1	AdaMerging	76.0	77.2	77.5	77.7	80.1
Surgery	70.0	71.1	70.9	73.4	73.2	73.6	Surgery	72.8	77.3	78.4	78.9	80.9
SyMerge	73.2	75.3	75.7	77.1	80.0	82.1	SyMerge	78.1	84.5	86.7	88.0	90.1

Table D. **Per-task performance for the ablation that removes classifier adaptation.** We merge ViT-B/32 models on 8 tasks and evaluate the merged encoder from **SyMerge** while using the original zero-shot classifier for each task.

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	Avg.
Task Arithmetic encoder + zero-shot head	55.2	54.9	66.7	78.9	80.2	69.7	97.3	50.4	69.1
AdaMerging encoder + zero-shot head	64.5	68.1	79.2	93.8	87.0	91.9	97.5	59.1	80.1
SyMerge encoder + zero-shot head	71.4	73.5	87.4	90.6	92.6	81.3	96.1	66.1	82.4

Online Adaptation. First, we simulate a streaming setting where the model receives unlabeled test samples sequentially (batch size of 1). For each incoming sample, the model updates its parameters and is immediately evaluated on that sample, prohibiting multiple passes. Table Ca compares the cumulative average performance of **SyMerge** against leading test-time adaptation methods on ViT-B/32. **SyMerge** demonstrates remarkable data efficiency. Specifically, in terms of *rapid adaptation*, with only 10 samples seen, **SyMerge** achieves 73.2% accuracy, surpassing AdaMerging and Surgery by over 3.0%p. Furthermore, regarding *sustained improvement*, as the data stream continues to 500 samples, **SyMerge** continuously improves to 82.1%, establishing a significant margin ($\sim 8.0\%$ p) over the baselines, which tend to plateau early.

Limited-Data Efficiency. We further evaluated performance using varying amounts of unlabeled test data (ranging from 1% to 100% of the test set). As shown in Table Cb, **SyMerge** consistently outperforms optimization-based TTA methods (AdaMerging, Surgery) as well as training-free merging baselines such as Task Arithmetic (69.1%), Ties Merging (72.9%), and LiNeS (74.1%). Notably, even with extremely limited data (1%), **SyMerge** achieves 78.1% accuracy, significantly outperforming the baselines. This result confirms that our expert-guided self-labeling strategy creates a reliable learning signal even from minimal data, making **SyMerge** highly suitable for practical scenarios where data collection is expensive or restricted.

C.6. Per-task results for improvement beyond classifier adaptation

To isolate whether **SyMerge** improves the merged encoder itself rather than only the task-specific layer, we run an ablation that removes classifier adaptation at evaluation time. We take the merged encoder produced by **SyMerge** on CLIP (Radford et al., 2021) ViT-B/32 for 8 vision tasks, replace each task head with the corresponding original zero-shot classifier, and evaluate without any further training. We compare this against the standard Task Arithmetic merged encoder paired with the same zero-shot classifiers. Table D reports per-task accuracies. Consistent with the main text, **SyMerge** improves the average accuracy from 69.1 to 82.4, showing that the gains largely come from better merged representations in the encoder, even when the classifier is kept fixed. Notably, this intrinsic improvement is also competitive with, and even exceeds, AdaMerging (80.1), which focuses on optimizing only the encoder merging coefficients without training task heads.

D. Proof for Proposition 3.1

We provide a proof for Proposition 3.1 in the main paper.

Proposition D.1. Assume cross-task linearity (Zhou et al., 2024) so that $f(x; \frac{1}{2}(\theta_i + \theta_j)) \approx \frac{1}{2}f(x; \theta_i) + \frac{1}{2}f(x; \theta_j)$, and suppose the loss function is convex in its output. Then the merged model $f_{\text{merge}}(x) = f(x; \frac{1}{2}(\theta_i + \theta_j))$ has an expected loss below the case of weight disentanglement.

Proof. Let $f_i(x) \triangleq f(x; \theta_i)$ and $f_j(x) \triangleq f(x; \theta_j)$. The expected loss on task j is denoted by $\mathcal{L}_j(f)$. We assume Cross-Task Linearity (CTL), so $f(x; \frac{1}{2}(\theta_i + \theta_j)) \approx \frac{1}{2}f_i(x) + \frac{1}{2}f_j(x)$. Given that the loss function L_j is convex with respect to its

Table E. Performance details of merged ViT-B/32 models when training different layers with merging coefficients for task-specific adjustments. We observe that consistently high performance can be achieved regardless of which task-specific single layer is trained.

Layer	SUN	Cars	RES.	Euro	SVH.	GTS.	MNI.	DTD	Avg.
1	73.7	78.0	95.4	99.9	96.3	98.7	99.5	79.1	90.1
2	74.3	78.2	95.8	99.9	96.1	98.7	99.5	79.0	90.2
3	74.4	78.5	95.9	99.9	96.5	98.8	99.5	79.3	90.4
4	74.5	78.4	95.9	99.8	96.7	98.7	99.6	79.2	90.4
5	74.6	78.4	96.0	99.9	96.5	99.0	99.6	79.1	90.4
6	74.6	78.4	95.2	99.8	96.2	98.9	99.1	79.5	90.2
7	74.7	78.1	95.6	99.3	96.4	98.7	99.3	79.8	90.2
8	74.8	77.9	95.9	99.6	96.2	98.1	99.1	79.5	90.1
9	75.0	78.7	95.8	99.9	96.4	98.6	99.5	79.5	90.4
10	74.8	78.5	95.8	99.9	96.2	98.9	99.5	79.4	90.4
11	75.2	78.6	95.9	99.9	96.5	98.6	99.3	79.7	90.5
12	75.0	77.8	95.4	99.7	95.8	97.5	99.3	79.5	90.0
Classifier	74.3	79.3	94.8	99.0	95.7	98.5	99.2	80.2	90.1
Late	71.2	70.8	94.3	99.2	96.1	98.8	99.3	77.2	88.4
Early	71.7	77.2	95.1	99.9	95.6	98.7	97.7	79.5	89.4
All	68.4	72.1	92.7	93.9	87.7	98.5	98.3	78.0	86.2

output, Jensen’s inequality provides a general upper bound for the merged model’s loss:

$$\mathcal{L}_j(f_{\text{merge}}) \leq \frac{1}{2}\mathcal{L}_j(f_i) + \frac{1}{2}\mathcal{L}_j(f_j). \quad (1)$$

We now analyze this bound under two conditions for $\theta_i = \theta_0 + \tau_i$, where θ_0 is a pre-trained model and τ_i is the task vector for task i .

Case A: Weight Disentanglement. This condition assumes that the task vector τ_i does not affect performance on task j , i.e., $\mathcal{L}_j(f_i) = \mathcal{L}_j(f_0)$. Substituting this into (1) yields:

$$\mathcal{L}_j(f_{\text{merge}}) \leq \frac{1}{2}\mathcal{L}_j(f_0) + \frac{1}{2}\mathcal{L}_j(f_j). \quad (2)$$

Case B: Synergistic Effect. This condition assumes that τ_i improves performance on task j , implying $\mathcal{L}_j(f_i) < \mathcal{L}_j(f_0)$. We can write this as $\mathcal{L}_j(f_i) = \mathcal{L}_j(f_0) - \epsilon_{ij}$ for some $\epsilon_{ij} > 0$. This gives a tighter bound:

$$\mathcal{L}_j(f_{\text{merge}}) \leq \frac{1}{2}(\mathcal{L}_j(f_0) - \epsilon_{ij}) + \frac{1}{2}\mathcal{L}_j(f_j) = \left(\frac{1}{2}\mathcal{L}_j(f_0) + \frac{1}{2}\mathcal{L}_j(f_j)\right) - \frac{\epsilon_{ij}}{2}. \quad (3)$$

Comparing (2) and (3), the upper bound on the loss is strictly lower by $\frac{\epsilon_{ij}}{2}$ under the synergistic effect. This proves that merging synergistic task vectors yields a superior model. $\square$

E. More Empirical Analyses

E.1. Detailed Results on Task-specific Layers

Our proposed method, **SyMerge**, introduces the concept of incorporating a task-specific layer to balance shared and task-specific representations during model merging. This approach allows effective adaptation while maintaining efficiency to address task conflicts in multi-task learning. ViT-B/32 is employed and initialized using Task Arithmetic-merged weights across all layers. When a particular layer is marked as task-specific, it is split into unique versions for each task, resulting in eight task-specific layers. While these layers are trained exclusively on corresponding task data, the remaining layers update only the merging coefficients using data from all tasks. Table E presents detailed task-level results. ‘Early’ refers to layers 1 to 6-th, while ‘Late’ refers to layers 7 to 12-th. Training both the merging coefficients and task-specific layers achieves performance comparable to or slightly better than training merging coefficients alongside the classifier. These findings highlight the minimal impact of task-specific layer choice on overall performance, reaffirming the robustness of our approach. The results also show consistent average performance across layers, demonstrating that merging coefficients effectively balance representations across tasks, mitigating biases even when other layers remain frozen.

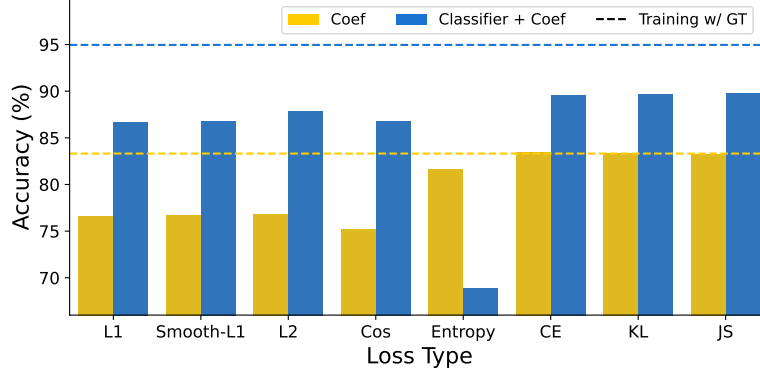


Figure C. **Comparison of training with various loss functions.** The yellow bars represent training only the coefficients, while the blue bars indicate the joint training of the classifier and coefficients. The dashed line corresponds to training with ground truth labels using cross-entropy loss, which is the upper bound.

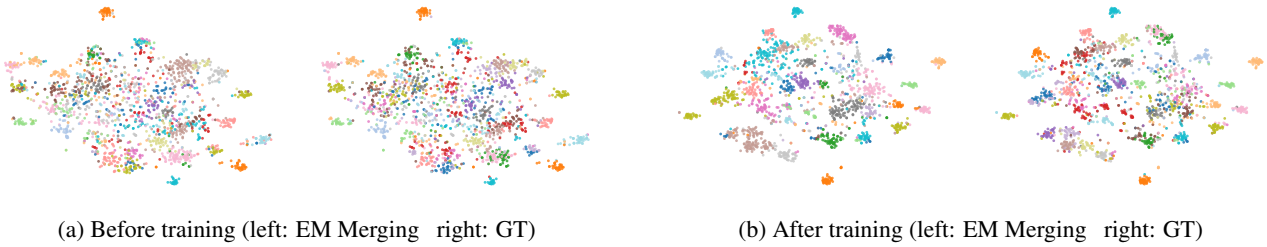


Figure D. **t-SNE visualization results on the DTD dataset before (a) and after (b) entropy minimization training.** The plots on the left use merged model predictions, while those on the right use ground truth labels. Training improves clustering, but prediction-ground truth mismatch remains.

E.2. On Loss Functions

In Figure C, we analyze the impact of various loss functions on aligning predictions between individual models and the merged model. By default, we employ cross-entropy with hard one-hot pseudo-labels in our main experiments for simplicity, while a soft-target KL-divergence variant yields comparable performance. When training only the coefficients, distance-based losses (e.g., L1, Smooth-L1, L2, and Cosine) demonstrate relatively lower performance compared to distribution-based losses (e.g., KL-divergence, JS-divergence, cross-entropy, and entropy). In contrast, when both the classifier and coefficients are trained jointly, most loss functions achieve strong performance under our method. However, entropy minimization, as explored in previous works (Tang et al., 2024a; Yang et al., 2024b), leads to a significant decline in performance during joint training.

Notably, in Table F, we observe that the entropy loss often fails to provide gradients in the correct direction due to its divergence from the ground truth (GT)-based loss. The table highlights that the cross-entropy loss, in contrast, aligns more closely with the GT, resulting in more accurate gradient directions. This highlights the importance of incorporating label information for guiding classifier training in the merged model.

To further support this point, Figure D shows the t-SNE visualization for the DTD dataset before and after entropy minimization training. While feature clustering improves after training, as shown in (b) compared to (a), a noticeable discrepancy remains between the predictions and the ground truth labels. These observations emphasize the importance of label guidance, even if incomplete, for achieving better alignment with the GT and improving the performance of the merged model.

Loss correlation. We conduct further experiments to validate whether the loss correlation shown in Figure 4 of the main paper remains consistent across different backbones (ViT-{B/32, L/14}). Table F shows the loss correlation results for 8 vision tasks using the merged ViT-B/32 and ViT-L/14 models. We observe a significant drop in correlation after training with entropy minimization. In contrast, the cross-entropy loss employed in our self-labeling approach maintains consistently high correlation across tasks. These findings are consistent with the results reported for ViT-B/32 in the main paper, suggesting

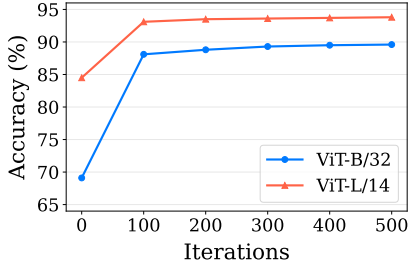
Table F. **Spearman correlation of losses with ground truth cross-entropy loss** for (a) ViT-B/32 and (b) ViT-L/14. Values closer to 1 indicate that the corresponding loss function exhibits better alignment with the loss computed using the ground truth.

		SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD
Entropy	Before	0.66	0.72	0.80	0.86	0.92	0.92	1.00	0.56
	After	0.32	-0.47	0.79	0.94	0.83	0.41	0.96	0.47
	Δ	(-0.34)	(-1.19)	(-0.01)	(+0.08)	(-0.09)	(-0.51)	(-0.04)	(-0.09)
Ours	Before	0.80	0.86	0.97	1.00	0.98	0.99	1.00	0.84
	After	0.82	0.88	0.99	1.00	0.99	1.00	1.00	0.88
	Δ	(+0.02)	(+0.02)	(+0.02)	(+0.00)	(+0.01)	(+0.01)	(+0.00)	(+0.04)

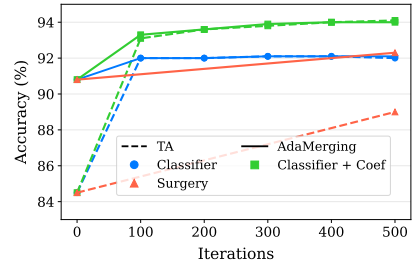
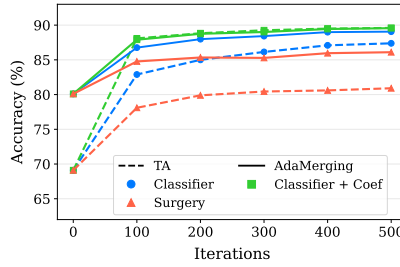
(a) ViT-B/32

		SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD
Entropy	Before	0.85	0.93	0.95	0.98	0.96	0.96	1.00	0.79
	After	0.63	0.76	0.94	0.81	0.79	0.95	0.82	0.83
	Δ	(-0.22)	(-0.17)	(-0.01)	(-0.17)	(-0.17)	(-0.01)	(-0.18)	(+0.04)
Ours	Before	0.87	0.96	0.98	1.00	0.98	1.00	1.00	0.89
	After	0.89	0.97	1.00	1.00	0.99	1.00	1.00	0.91
	Δ	(+0.02)	(+0.01)	(+0.02)	(+0.00)	(+0.01)	(+0.00)	(+0.00)	(+0.02)

(b) ViT-L/14

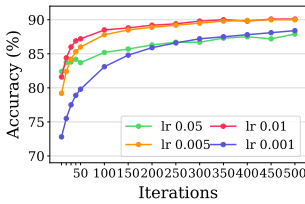


(a) Learning curves across models.

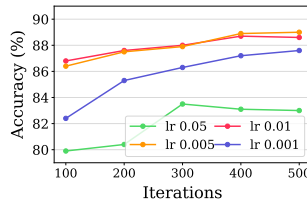


(b) Impact of initialization on ViT-B/32 and ViT-L/14.

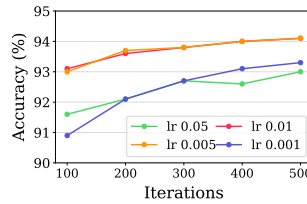
Figure E. **Learning curves and analysis.** (a) Comparison of average accuracy for ViT-B/32 and ViT-L/14 across training iterations. (b) Impact of initializations on ViT-B/32 (left) and ViT-L/14 (right), comparing fixed-value initialization from Task Arithmetic (dashed lines) and learned-value one in AdaMerging (solid lines).



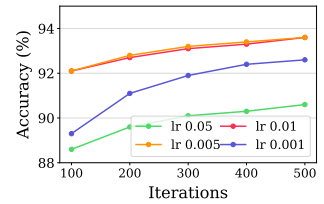
(a) 8 tasks on ViT-B/32.



(b) 20 tasks on ViT-B/32.



(c) 8 tasks on ViT-L/14.



(d) 20 tasks on ViT-L/14.

Figure F. **Impact of learning rate on performance.** Average accuracy curves across training iterations for ViT-B/32 and ViT-L/14 under varying learning rates. We report results for both 8-task and 20-task settings to analyze the hyperparameter sensitivity.

that our self-labeling approach with the cross-entropy loss could be more effective than the entropy minimization loss.

E.3. Convergence Analysis

Figure E illustrates the learning curve of **SyMerge** applied to the 8 vision tasks over training iterations. In Figure Ea, we compare the performance of ViT-B/32 and ViT-L/14 models. The results demonstrate that our method achieves near-optimal performance within the first 100 iterations and converges quickly, regardless of model size. Notably, larger models like ViT-L/14 consistently achieve higher final accuracies, followed by ViT-B/32, which highlights the advantages of model capacity in capturing task-specific knowledge more effectively. Figure Eb shows that when training a task-specific adapter

Table G. Empirical Verification of Cross-Task Linearity (CTL). We measure the cosine similarity between the output of a parameter-interpolated model $f(x; \theta(\alpha))$ and the output interpolation $(1 - \alpha)f(x; \theta_{\text{src}}) + \alpha f(x; \theta_{\text{tgt}})$ on ViT-B/32. Using SUN397 as the source task (θ_{src}), the consistently high similarity scores across all coefficients α (even at $\alpha = 0.5$) confirm that the CTL assumption holds robustly in our experimental setting.

Task	Interpolation Coefficient (α)								
	0.1	0.2	0.3	0.4	0.5	0.6	0.7	0.8	0.9
Cars	1.00	0.99	0.98	0.97	0.97	0.97	0.98	0.99	1.00
RESISC45	0.99	0.98	0.97	0.96	0.95	0.95	0.96	0.98	0.99
EuroSAT	0.99	0.98	0.96	0.95	0.94	0.94	0.95	0.97	0.99
SVHN	0.99	0.97	0.94	0.90	0.87	0.85	0.87	0.92	0.97
GTSRB	0.99	0.96	0.93	0.90	0.88	0.88	0.90	0.94	0.98
MNIST	0.99	0.97	0.94	0.92	0.90	0.89	0.91	0.95	0.98
DTD	0.99	0.98	0.96	0.96	0.96	0.96	0.97	0.99	1.00

Table H. Ablation study on confidence-based filtering. We report the mean $\pm$ standard deviation over 5 runs. Applying the filtering consistently improves performance across different numbers of tasks and model scales.

Setting	Model	Accuracy (%)	
		w/o Filtering	w/ Filtering
8 tasks	ViT-B/32	89.7 $\pm$ 0.1	90.1 $\pm$ 0.1 (+0.4)
14 tasks	ViT-B/32	88.3 $\pm$ 0.2	88.7 $\pm$ 0.1 (+0.4)
20 tasks	ViT-B/32	88.5 $\pm$ 0.3	88.6 $\pm$ 0.4 (+0.1)
8 tasks	ViT-L/14	93.9 $\pm$ 0.0	94.1 $\pm$ 0.0 (+0.2)

only, the Surgery model is sensitive to initialization. In contrast, our method jointly optimizes the merging coefficient and task-specific layer, making it robust to initialization sensitivity. Figure F shows the convergence of ViT-B/32 and ViT-L/14 under various learning rates for 8-task and 20-task settings. Excluding the extreme learning rate case, our method exhibits rapid convergence and robust performance across most settings. These observations demonstrate the robustness and efficiency of **SyMerge** across different model scales and hyperparameter settings.

E.4. Empirical Verification of Cross-Task Linearity

A key theoretical assumption in our work is Cross-Task Linearity (CTL) (Zhou et al., 2024), which posits that the representation of a merged model can be approximated by the linear interpolation of individual model representations. To rigorously validate this assumption, we conducted a quantitative analysis comparing the two interpolation schemes using ViT-B/32. Specifically, we define the parameter-space interpolation as $\theta(\alpha) = (1 - \alpha)\theta_{\text{src}} + \alpha\theta_{\text{tgt}}$ and the representation-space interpolation as $F_{\text{rep}}(\alpha) = (1 - \alpha)f(x; \theta_{\text{src}}) + \alpha f(x; \theta_{\text{tgt}})$. If CTL holds strictly, the output of the merged model $f(x; \theta(\alpha))$ should align closely with $F_{\text{rep}}(\alpha)$. We selected SUN397 as the source task (θ_{src}) and paired it with seven other target tasks (θ_{tgt}) to measure the Cosine Similarity between $f(x; \theta(\alpha))$ and $F_{\text{rep}}(\alpha)$ across varying interpolation coefficients $\alpha \in [0.1, 0.9]$.

Table G presents the results. We observe that the cosine similarity remains remarkably high across all task pairs and α values. Most notably, even at the interpolation midpoint ($\alpha = 0.5$), the similarity scores remain consistently high. These findings empirically demonstrate that within the shared solution basin derived from a common pre-trained model, the encoder’s behavior exhibits strong linearity. Consequently, this confirms that the CTL assumption serves as a robust approximation in our experimental regime, supporting the validity of our theoretical proposition.

E.5. Impact of Confidence-based Filtering

To verify the effect of confidence-based filtering, we evaluate the model without it. As detailed in Table H, employing the filtering strategy yields consistent but modest performance gains across varying task counts and model scales (e.g., +0.4%p on 8 tasks). Crucially, even without filtering, **SyMerge** maintains state-of-the-art performance (e.g., 89.7% on 8 tasks), significantly outperforming existing baselines. This confirms that while filtering serves as an effective safeguard against spurious supervision from potentially noisy experts, the core efficacy of **SyMerge** stems from the synergistic joint optimization framework itself.

Table I. Computational efficiency. We report the peak GPU memory usage and the number of trainable parameters during adaptation on 8 vision tasks. To prevent OOM errors on baselines, all methods are evaluated under a standardized sequential update setting (batch size=16). **SyMerge** achieves high efficiency, comparable to the most lightweight baseline.

Method	Peak GPU Mem (GB)		Params (M)	
	ViT-B/32	ViT-L/14	ViT-B/32	ViT-L/14
AdaMerging (Yang et al., 2024b)	7.09	22.88	0.00	0.00
Surgery (Yang et al., 2024a)	7.13	30.44	0.13	0.20
WEMoE (Tang et al., 2024a)	8.01	33.57	7.16	25.39
SyMerge	7.12	23.71	0.39	0.59

Table J. Per-step runtime comparison on ViT-B/32. Runtime is measured in milliseconds. SyMerge uses on-the-fly expert supervision, while SyMerge[†] uses cached expert predictions.

# Tasks	AdaMerging	Surgery	WEMoE	SyMerge	SyMerge [†]
2	71	170	331	170	65
4	87	183	331	179	80
8	114	211	428	215	111
14	162	259	743	262	153
20	207	299	1047	303	195

E.6. Computational Efficiency Analysis

To evaluate the resource efficiency of **SyMerge**, we measured the peak GPU memory usage and the number of trainable parameters across 8 vision tasks. For a fair comparison, we standardized all methods to employ sequential updates (task-by-task optimization) with a batch size of 16. This adaptation was necessary because the original simultaneous update schemes of baselines like AdaMerging and WEMoE incur excessive memory costs, often leading to out-of-memory (OOM) errors on large backbones (*e.g.*, ViT-L/14).

As shown in Table I, **SyMerge** demonstrates remarkable efficiency. Its peak memory usage is comparable to AdaMerging, the most lightweight baseline, and significantly lower than WEMoE. Furthermore, **SyMerge** introduces negligible trainable parameters (*e.g.*, <0.6M for ViT-L/14). This confirms that our method achieves state-of-the-art performance with a minimal computational footprint, making it practical for resource-constrained environments.

We further analyze the runtime overhead of expert supervision in Table J. In our default implementation, expert predictions are generated on-the-fly during adaptation. Since only the expert corresponding to the current task is queried at each step, the per-step cost does not scale with the total number of experts. Across different numbers of tasks, **SyMerge** has runtime comparable to Surgery and is substantially faster than WEMoE. In addition, when expert predictions are cached, the runtime is further reduced and becomes close to AdaMerging. These results show that the additional cost of expert supervision is manageable and can be largely mitigated by caching.

E.7. Sparsity Visualization

Inspired by the literature (Yu et al., 2024; Yadav et al., 2023; Davari & Belilovsky, 2024), which highlights how redundant parameters degrade performance due to conflicts among task-specific parameters, we investigate this phenomenon in our method. We analyze the learned merging coefficients to observe their impact, specifically exploring *sparsity* after training them alone and jointly with the classifier.

To confirm this trend across various backbones, Figure G and H provide the layer-wise merging coefficients for ViT-B/32 and ViT-L/14. All models are optimized using the cross-entropy loss with the predictions of individual models as guidance. In these figures, (a) represents training only the coefficients, while (b) includes joint training of the classifier and coefficients.

We find that jointly training a task-specific layer, like the classifier, with merging coefficients increases the proportion of coefficients concentrated near zero (*i.e.*, smaller than $1e-5$), leading to improved accuracy. For example, joint training boosts the share of near-zero coefficients from 37.2% to 55.9%, with a corresponding accuracy improvement from 84.6% to 90.1%. This trend is consistent across backbones like ViT-L/14, where the proportion of sparse coefficients similarly rises (from 23.3% to 55.0%) as does the average accuracy (from 91.5% to 94.1%). These results indicate that training the classifier

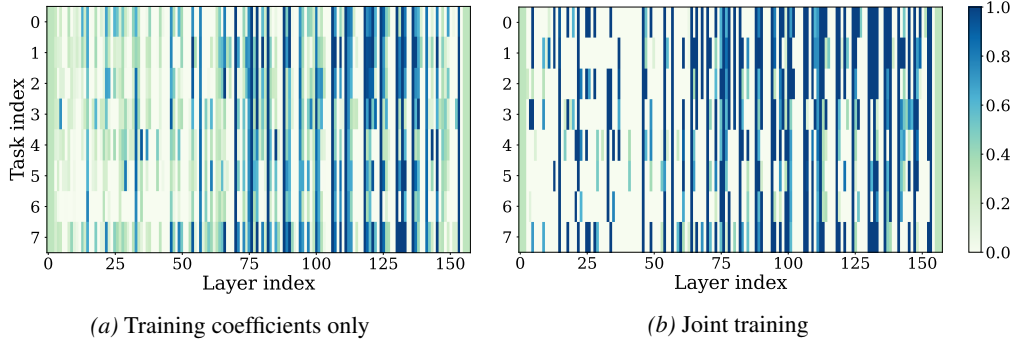


Figure G. Impact of joint training on merging coefficient sparsity on ViT-B/32.

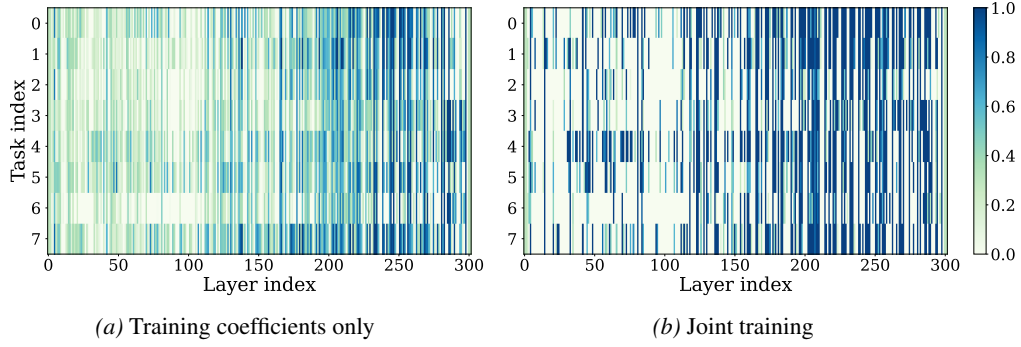


Figure H. Impact of joint training on merging coefficient sparsity on ViT-L/14.

complements sparsity in the merging coefficients, effectively reducing task conflicts by pruning unnecessary parameters and enhancing performance.

Table K. Ablations of the corrupted test dataset on ViT-B/32.

Method	Cars	EuroSAT	RESISC45	GTSRB	Avg.	Cars	EuroSAT	RESISC45	GTSRB	Avg.
Original Test Set						Corrupted Test Set (Motion Blur)				
Individual	77.7	99.9	96.1	98.7	93.1	75.1	75.2	94.1	95.3	84.9
Task Arithmetic	67.0	94.0	82.6	75.1	79.7	63.8	55.1	78.0	53.0	62.5
Ties-Merging	67.5	83.7	79.3	65.4	74.0	65.6	52.3	74.0	44.9	59.2
PCB-Merging	70.8	89.6	84.6	80.7	81.4	68.4	57.0	79.5	60.2	66.3
LiNeS	69.5	96.3	84.6	82.4	83.2	67.0	61.1	80.4	61.3	67.5
Consensus TA	67.1	95.4	78.0	84.9	81.4	63.9	59.5	56.2	80.3	65.0
Iso-CTS	73.6	96.3	91.3	93.4	88.6	69.7	67.9	76.4	87.8	75.5
AdaMerging	76.2	96.0	87.5	97.0	89.2	71.7	69.4	83.3	91.9	79.1
Surgery w/ Ada	73.4	98.5	91.2	97.8	90.2	69.9	73.7	88.7	90.1	80.6
WEMoE	79.1	99.3	95.5	99.1	93.2	76.6	54.5	93.4	96.1	80.2
SyMerge	77.6	99.4	95.3	98.3	92.7	74.7	76.0	93.2	93.6	84.4
Corrupted Test Set (Impulse Noise)						Corrupted Test Set (Gaussian Noise)				
Individual	69.1	14.3	79.0	66.9	57.3	73.9	14.0	90.1	74.6	63.2
Task Arithmetic	58.2	16.3	56.9	24.8	39.0	62.8	26.3	68.5	36.5	48.5
Ties-Merging	59.7	13.7	55.0	20.9	37.3	63.6	23.3	66.6	29.3	45.7
PCB-Merging	62.3	15.7	61.0	28.4	41.8	66.5	26.7	73.0	41.4	51.9
LiNeS	60.6	18.5	58.3	28.1	41.4	65.7	28.9	70.2	40.7	51.4
Consensus TA	58.2	17.9	24.4	56.9	39.4	62.9	29.0	36.8	69.4	49.5
Iso-CTS	62.9	20.4	33.5	64.2	45.2	69.0	33.5	46.5	77.1	56.5
AdaMerging	62.7	14.3	65.5	53.8	49.0	69.6	17.4	73.9	63.1	56.0
Surgery w/ Ada	62.7	12.4	68.3	48.8	48.1	68.6	11.9	82.3	59.7	55.6
WEMoE	70.3	11.6	83.5	9.6	43.7	74.9	11.6	90.6	8.0	46.2
SyMerge	69.0	13.1	79.5	63.6	56.3	74.4	12.6	88.8	72.4	62.1
Corrupted Test Set (Pixelate)						Corrupted Test Set (Spatter)				
Individual	76.7	96.7	95.6	96.0	91.3	71.1	88.5	83.6	94.8	84.5
Task Arithmetic	65.8	78.5	81.1	55.9	70.3	57.5	55.9	67.2	56.8	59.3
Ties-Merging	66.8	68.7	78.9	46.3	65.2	58.2	45.5	64.6	46.5	53.7
PCB-Merging	69.5	75.4	84.0	62.1	72.8	61.3	52.9	70.5	63.0	61.9
LiNeS	68.4	84.7	83.3	61.7	74.5	59.8	59.6	68.8	64.8	63.3
Consensus TA	65.9	82.6	58.5	83.5	72.6	57.0	65.4	58.4	68.6	62.4
Iso-CTS	71.7	82.0	76.2	90.4	80.1	64.1	71.8	76.0	74.2	71.5
AdaMerging	71.2	89.8	81.7	90.8	83.4	66.5	14.6	74.5	92.0	61.9
Surgery w/ Ada	71.8	94.9	90.0	89.2	86.5	63.1	87.0	74.7	90.5	78.8
WEMoE	77.6	96.5	94.5	96.0	91.1	72.0	11.6	86.0	95.1	66.2
SyMerge	76.4	96.5	95.1	95.4	90.8	71.2	87.9	82.9	94.4	84.1
Corrupted Test Set (Contrast)						Corrupted Test Set (JPEG Compression)				
Individual	68.1	22.2	56.7	82.6	57.4	77.2	67.4	95.4	92.9	83.2
Task Arithmetic	55.2	28.0	48.2	48.5	45.0	65.9	65.9	80.3	53.3	66.4
Ties-Merging	57.5	34.3	52.4	42.5	46.6	66.8	51.4	78.1	43.1	59.8
PCB-Merging	58.6	33.7	53.3	56.4	50.5	70.3	60.7	83.5	59.1	68.4
LiNeS	58.2	29.4	49.0	55.3	48.0	68.9	68.7	82.5	59.9	70.0
Consensus TA	55.7	31.1	47.9	49.9	46.2	66.5	65.4	53.7	82.7	67.1
Iso-CTS	63.0	26.6	69.6	55.0	53.5	72.5	57.8	71.1	89.8	72.8
AdaMerging	65.8	39.7	60.2	90.7	64.1	72.1	65.4	82.9	87.3	76.9
Surgery w/ Ada	62.0	21.1	50.6	83.0	54.2	72.1	66.0	89.8	84.9	78.2
WEMoE	70.9	47.0	65.7	93.1	69.2	78.0	70.5	94.3	92.8	83.9
SyMerge	67.6	20.5	56.5	81.3	56.5	77.3	65.0	94.3	91.5	82.0

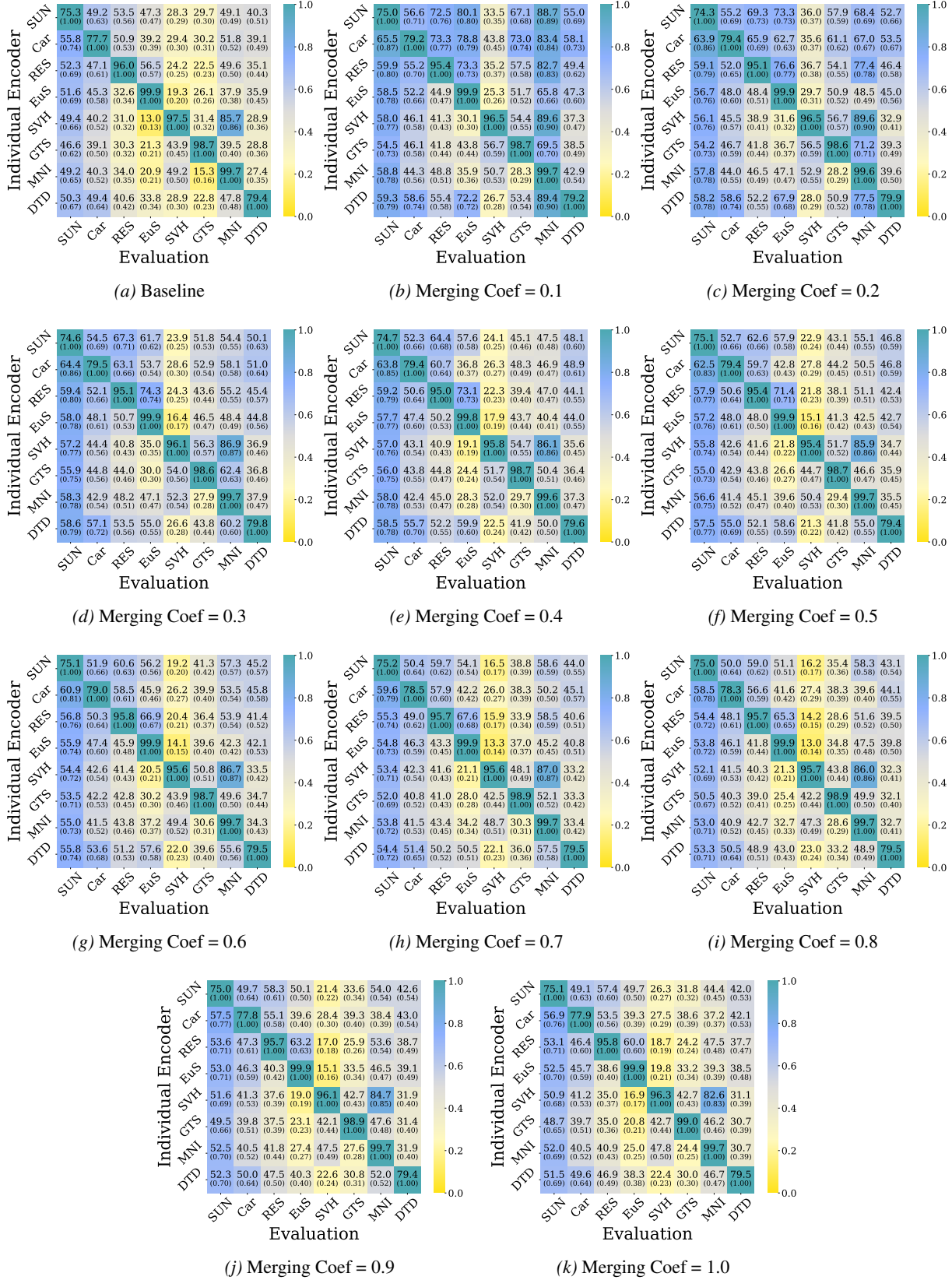


Figure I. Additional results for cross-task evaluations across diverse encoders and heads from different tasks.

Table L. Multi-task performance when merging ViT-B/32 models on 8 tasks

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	Avg.
Individual	75.3	77.7	96.1	99.7	97.5	98.7	99.7	79.4	90.5
Task Arithmetic (Ilharco et al., 2023)	55.2	54.9	66.7	78.9	80.2	69.7	97.3	50.4	69.1
Ties Merging (Yadav et al., 2023)	65.0	64.4	74.8	77.4	81.2	69.3	96.5	54.5	72.9
PCB Merging (Du et al., 2024)	62.9	62.0	77.1	80.1	87.5	78.5	98.7	58.4	75.6
LiNeS w/ TA (Wang et al., 2025)	63.7	63.9	75.1	86.1	79.4	72.2	96.2	56.5	74.1
Consensus TA (Wang et al., 2024)	62.9	61.0	71.0	82.7	86.8	79.3	98.1	57.2	74.9
TSV-M (Gargiulo et al., 2025)	68.8	72.0	85.9	94.7	90.9	91.2	99.2	69.1	84.0
ISO-CTS (Marczak et al., 2025)	70.3	73.7	89.1	95.0	86.3	90.9	99.0	69.4	84.2
EMR-Merging (Huang et al., 2024a)	75.2	72.8	93.5	99.5	96.9	98.1	99.6	74.4	88.7
AdaMerging (Yang et al., 2024b)	64.5	68.1	79.2	93.8	87.0	91.9	97.5	59.1	80.1
Surgery w/ Ada. (Yang et al., 2024a)	71.2	72.0	92.3	99.0	92.2	97.9	99.0	76.1	87.5
WeMoE (Tang et al., 2024a)	74.1	77.4	93.7	99.1	96.2	98.9	99.6	76.4	89.4
ProbSurgery w/ Ada. (Wei et al., 2025)	70.3	71.6	91.9	98.8	92.5	97.9	99.0	77.5	87.4
SyMerge	74.7	79.0	95.1	98.9	95.7	98.3	99.1	80.0	90.1

Table M. Multi-task performance when merging ViT-B/32 models on 14 tasks.

Method	SUN	Cars	RES.	Euro	SVH.	GTS.	MNI.	DTD	C100	FER	Flower	Pet	PCAM	STL	Avg.
Individual	75.3	77.7	96.1	99.7	97.5	98.7	99.7	79.4	89.1	72.5	90.4	91.8	87.8	97.8	89.5
Task Arithmetic (Ilharco et al., 2023)	63.9	59.5	67.5	67.7	52.9	47.0	80.8	48.2	69.6	42.9	67.6	87.5	63.2	96.7	65.4
Ties Merging (Yadav et al., 2023)	65.1	61.8	68.3	63.7	51.3	45.9	80.0	48.7	69.7	42.4	68.1	88.0	62.1	97.2	65.2
PCB Merging (Du et al., 2024)	52.5	40.7	62.2	71.3	60.8	61.7	94.1	50.3	51.1	44.0	57.4	82.0	75.1	90.4	63.8
LiNeS w/ TA (Wang et al., 2025)	64.3	59.9	69.6	75.4	58.8	54.6	85.1	51.1	70.0	45.0	69.3	88.0	64.3	96.8	68.0
Consensus TA (Wang et al., 2024)	63.8	57.5	69.5	77.9	69.2	60.4	93.7	52.4	67.3	44.4	68.9	88.1	74.6	95.9	70.3
TSV-M (Gargiulo et al., 2025)	66.7	62.0	81.5	93.6	84.6	84.8	98.7	65.9	71.0	63.3	76.4	91.0	84.9	97.1	80.1
ISO-CTS (Marczak et al., 2025)	68.9	66.1	85.7	91.8	80.5	85.3	98.4	67.3	74.9	59.3	82.0	89.0	81.8	97.4	80.6
EMR-Merging (Huang et al., 2024a)	70.4	68.3	92.5	99.0	96.1	97.6	99.5	72.0	81.2	68.5	85.9	90.7	86.7	97.3	86.1
AdaMerging (Yang et al., 2024b)	64.3	68.5	81.7	92.6	86.6	90.8	97.5	60.2	67.3	53.1	73.8	87.9	53.8	96.3	76.7
Surgery w/ Ada. (Yang et al., 2024a)	69.1	71.5	89.5	97.8	90.2	95.3	98.6	73.6	73.4	66.3	86.0	92.2	82.3	98.1	84.6
WeMoE (Tang et al., 2024a)	74.2	78.1	94.0	98.2	95.8	98.5	99.6	75.9	83.7	32.8	90.8	91.8	51.3	97.8	83.0
ProbSurgery w/ Ada. (Wei et al., 2025)	68.9	68.7	91.1	98.6	92.2	95.0	98.3	77.2	72.3	64.4	87.1	91.8	84.6	98.4	84.9
SyMerge	74.0	78.7	94.6	98.8	94.6	97.7	98.9	80.0	84.9	71.0	92.2	92.2	86.1	98.1	88.7

Table N. Multi-task performance when merging ViT-B/32 models on 20 tasks.

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	CIFAR100	FER2013
Individual	75.3	77.7	96.1	99.7	97.5	98.7	99.7	79.4	89.1	72.5
Task arithmetic (Ilharco et al., 2023)	61.8	53.0	61.9	57.5	49.8	44.6	77.9	45.6	65.4	41.4
Ties Merging (Yadav et al., 2023)	64.6	58.7	66.4	59.7	54.9	46.7	80.1	47.5	69.0	41.8
PCB Merging (Du et al., 2024)	41.3	21.5	44.6	47.4	52.4	41.9	86.9	39.7	40.7	38.5
LiNeS w/ TA (Wang et al., 2025)	63.6	55.3	66.3	65.0	56.7	51.3	81.6	48.8	68.3	44.1
Consensus TA (Wang et al., 2024)	63.7	53.4	66.3	63.8	63.5	52.2	89.5	49.4	66.3	41.1
TSV-M (Gargiulo et al., 2025)	65.8	54.1	77.8	89.8	76.9	74.9	94.1	60.6	69.5	58.3
ISO-CTS (Marczak et al., 2025)	67.0	56.2	80.4	83.2	75.4	76.5	96.4	63.9	74.0	57.0
EMR-Merging (Huang et al., 2024a)	71.0	67.6	91.1	98.6	94.4	96.7	99.4	71.0	81.1	65.7
AdaMerging (Yang et al., 2024b)	63.7	65.5	77.9	90.8	75.0	89.3	96.7	56.2	67.7	48.0
Surgery w/ Ada. (Yang et al., 2024a)	67.9	69.2	89.0	97.8	86.1	95.3	98.4	71.8	73.8	63.5
WeMoE (Tang et al., 2024a)	80.1	91.9	95.5	98.7	96.5	98.6	98.9	76.4	88.7	17.2
ProbSurgery w/ Ada. (Wei et al., 2025)	67.9	64.6	91.2	98.0	90.9	95.1	98.5	74.5	67.1	60.4
SyMerge	73.4	78.2	93.7	98.0	92.4	97.1	98.8	79.8	83.7	69.5

Method	Flowers	Pet	PCAM	STL10	CIFAR10	EMNIST	FMNIST	Food101	KMNIST	R-SST2
Individual	90.4	91.8	87.8	97.8	97.9	99.7	95.4	89.1	98.4	74.8
Task arithmetic	63.1	86.0	65.8	94.4	91.5	39.6	73.9	72.1	12.2	57.8
Ties Merging	66.4	87.4	63.8	96.4	92.7	41.2	73.2	79.5	12.5	60.4
PCB Merging	43.1	71.7	68.4	85.9	80.3	62.1	74.1	34.5	27.0	52.4
LiNeS w/ TA	67.0	87.4	64.7	96.0	92.5	45.5	75.4	78.0	13.8	53.2
Consensus TA	66.6	86.3	68.9	95.8	92.5	51.9	74.5	75.5	17.0	62.4
TSV-M	72.7	90.1	83.6	96.8	94.2	94.4	84.0	79.3	44.6	70.4
ISO-CTS	77.6	89.5	82.9	97.0	95.1	83.9	85.7	77.4	49.6	70.3
EMR-Merging	83.8	91.3	85.9	97.7	96.8	99.6	93.2	83.7	91.4	71.3
AdaMerging	68.0	87.6	54.1	96.5	91.3	30.8	80.9	79.7	12.6	60.2
Surgery w/ Ada.	83.6	91.7	83.7	98.2	94.6	97.0	88.9	83.9	82.1	73.8
WeMoE	98.3	96.0	51.2	99.4	98.0	99.1	37.4	94.8	10.0	49.9
ProbSurgery w/ Ada.	84.1	91.6	84.4	98.1	92.6	98.4	85.1	81.5	92.8	73.6
SyMerge	91.7	92.2	85.6	97.8	95.5	98.2	90.5	85.3	95.4	76.1

Table O. Multi-task performance when merging ViT-L/14 models on 8 tasks.

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	Avg.
Individual	82.3	92.4	97.4	100	98.1	99.2	99.7	84.1	94.2
Task Arithmetic (Ilharco et al., 2023)	73.9	82.1	86.6	94.1	87.9	86.7	98.9	65.6	84.5
Ties Merging (Yadav et al., 2023)	76.5	85.0	89.3	95.7	90.3	83.3	99.0	68.8	86.0
PCB Merging (Du et al., 2024)	75.9	85.9	89.6	95.7	89.9	92.3	99.2	71.4	87.5
LiNeS w/ TA (Wang et al., 2025)	74.5	85.4	88.8	95.4	90.8	90.8	99.3	70.4	86.9
Consensus TA (Wang et al., 2024)	74.9	83.0	88.2	95.4	91.3	91.5	99.1	69.6	86.6
TSV-M (Gargiulo et al., 2025)	78.4	89.9	93.7	98.7	95.6	96.5	99.5	79.9	91.5
ISO-CTS (Marczak et al., 2025)	80.6	91.7	96.0	99.2	95.1	98.5	99.5	83.0	93.0
EMR-Merging (Huang et al., 2024a)	83.2	90.7	96.8	99.7	97.9	99.1	99.7	82.7	93.7
AdaMerging (Yang et al., 2024b)	79.0	90.3	90.8	96.2	93.4	98.0	99.0	79.9	90.8
Surgery w/ Ada. (Yang et al., 2024a)	80.3	90.8	94.3	98.2	94.1	98.7	99.2	82.5	92.3
WeMoE (Tang et al., 2024a)	81.4	92.6	95.4	99.4	97.7	99.3	99.7	83.7	93.6
ProbSurgery w/ Ada. (Wei et al., 2025)	79.1	91.1	95.5	99.1	95.2	99.0	99.3	83.4	92.7
SyMerge	81.8	92.6	97.2	99.7	97.8	99.1	99.5	84.8	94.1

Table P. Multi-task performance when merging ViT-L/14 models on 14 tasks.

Method	SUN	Cars	RES.	Euro	SVH.	GTS.	MNI.	DTD	C100	FER	Flower	Pet	PCAM	STL	Avg.
Individual	82.3	92.4	97.4	100	98.1	99.2	99.7	84.1	93.3	76.5	97.9	95.2	90.0	99.3	93.2
Task Arithmetic (Ilharco et al., 2023)	71.6	79.0	80.6	86.5	75.6	65.8	96.0	60.9	83.9	43.5	80.2	94.9	79.7	99.4	78.4
Ties Merging (Yadav et al., 2023)	73.3	78.1	84.3	87.6	84.4	79.7	98.5	63.4	81.9	47.5	78.2	94.7	76.9	99.2	80.5
PCB Merging (Du et al., 2024)	72.2	73.3	82.4	89.3	86.2	85.2	98.9	63.6	76.8	47.7	87.4	94.4	82.4	98.7	81.3
LiNeS w/ TA (Wang et al., 2025)	72.6	79.7	83.1	89.9	78.2	72.6	97.2	63.6	84.5	45.7	81.1	95.6	82.2	99.4	80.4
Consensus TA (Wang et al., 2024)	73.6	79.2	85.3	94.3	86.0	82.8	98.6	63.7	82.7	47.9	78.4	94.9	86.1	99.1	82.3
TSV-M (Gargiulo et al., 2025)	76.3	84.5	92.2	97.7	93.7	94.1	99.5	75.2	85.8	66.7	87.5	95.9	85.8	99.6	88.2
ISO-CTS (Marczak et al., 2025)	79.2	88.6	95.1	98.7	91.9	96.9	99.4	80.6	88.2	62.1	96.5	96.1	82.5	99.5	89.7
EMR-Merging (Huang et al., 2024a)	80.5	89.4	96.5	99.6	97.7	99.0	99.7	81.3	90.3	71.3	92.9	95.1	86.4	99.4	91.4
AdaMerging (Yang et al., 2024b)	76.0	91.1	92.1	97.0	93.5	97.4	98.9	77.4	82.5	50.8	89.9	95.7	51.2	99.2	85.2
Surgery w/ Ada. (Yang et al., 2024a)	77.9	91.2	94.7	98.4	94.4	98.2	99.3	81.5	84.6	71.4	95.4	95.9	83.7	99.5	90.4
WEMoE (Tang et al., 2024a)	80.7	92.6	95.3	98.8	96.7	98.6	99.7	83.2	91.4	44.7	98.6	96.2	62.2	99.3	88.4
ProbSurgery w/ Ada. (Wei et al., 2025)	77.5	90.9	95.1	98.7	94.6	98.7	99.3	81.4	85.0	71.6	96.7	96.1	85.3	99.5	90.7
SyMerge	81.2	92.0	96.8	99.7	97.0	99.0	99.4	84.8	91.3	76.2	98.5	95.7	88.7	99.5	92.8

Table Q. Multi-task performance when merging ViT-L/14 models on 20 tasks.

Method	SUN397	Cars	RESISC45	EuroSAT	SVHN	GTSRB	MNIST	DTD	CIFAR100	FER2013
Individual	82.3	92.4	97.4	100.0	98.1	99.2	99.7	84.1	93.3	76.5
Task arithmetic (Ilharco et al., 2023)	71.2	76.2	78.0	79.3	75.3	65.7	95.9	59.7	82.5	42.0
Ties Merging (Yadav et al., 2023)	69.7	63.2	71.7	63.3	81.1	64.1	97.2	56.5	71.8	43.5
PCB Merging (Du et al., 2024)	68.2	56.5	70.2	68.4	81.3	68.1	97.6	56.5	68.1	43.8
LiNeS w/ TA (Wang et al., 2025)	72.1	77.6	80.6	84.6	76.5	69.0	96.5	62.0	83.3	43.5
Consensus TA (Wang et al., 2024)	73.7	75.9	83.2	84.8	82.2	71.8	97.7	61.8	82.1	45.5
TSV-M (Gargiulo et al., 2025)	74.9	79.7	89.9	95.7	90.3	90.4	97.9	72.8	83.3	60.1
ISO-CTS (Marczak et al., 2025)	77.6	82.1	93.7	98.5	90.4	96.4	99.0	78.9	86.4	59.5
EMR-Merging (Huang et al., 2024a)	79.5	88.9	95.9	99.3	96.9	98.7	99.6	79.3	90.1	64.6
AdaMerging (Yang et al., 2024b)	75.2	90.4	91.5	96.5	88.1	97.0	96.2	71.7	80.9	49.5
Surgery w/ Ada. (Yang et al., 2024a)	76.8	90.7	94.0	98.0	90.8	98.1	98.5	78.3	83.0	69.3
WEMoE (Tang et al., 2024a)	80.1	91.9	95.5	98.7	96.5	98.6	98.9	76.4	88.7	17.2
ProbSurgery w/ Ada. (Wei et al., 2025)	76.9	89.7	94.4	98.6	92.5	98.6	98.7	79.7	83.4	69.8
SyMerge	80.9	91.8	96.5	99.1	96.4	98.4	99.4	84.9	91.0	75.7

Method	Flowers	Pet	PCAM	STL10	CIFAR10	EMNIST	FMNIST	Food101	KMNIST	R-SST2
Individual	97.9	95.2	90.0	99.3	99.3	99.8	96.0	95.4	98.7	85.6
Task arithmetic	71.2	76.2	78.0	79.3	75.3	65.7	95.9	59.7	82.5	42.0
Ties Merging	67.5	93.7	70.3	97.4	94.8	85.7	83.5	82.0	34.9	67.1
PCB Merging	74.0	93.2	75.9	97.0	93.8	85.9	83.9	78.2	37.4	69.5
LiNeS w/ TA	72.1	77.6	80.6	84.6	76.5	69.0	96.5	62.0	83.3	43.5
Consensus TA	76.4	95.1	79.4	99.0	97.5	85.2	85.4	91.3	38.6	71.2
TSV-M	85.3	96.0	86.0	99.4	98.1	98.8	90.9	92.4	79.7	82.8
ISO-CTS	94.1	95.4	82.8	99.4	98.3	95.6	92.3	93.2	89.2	82.3
EMR-Merging	96.5	95.7	87.1	99.5	99.0	99.8	94.5	94.1	95.1	85.1
AdaMerging	88.4	95.5	50.4	99.0	97.0	97.0	91.5	92.1	10.0	83.9
Surgery w/ Ada.	94.2	95.7	83.3	99.4	97.9	99.1	92.3	92.4	72.7	85.6
WEMoE	98.3	96.0	51.2	99.4	98.0	99.1	37.4	94.8	10.0	49.9
ProbSurgery w/ Ada.	95.8	95.1	83.9	99.4	98.1	99.2	92.7	92.8	80.3	85.0
SyMerge	98.4	95.7	88.0	99.4	98.6	99.2	93.7	93.8	97.9	85.5